

Digestive tract

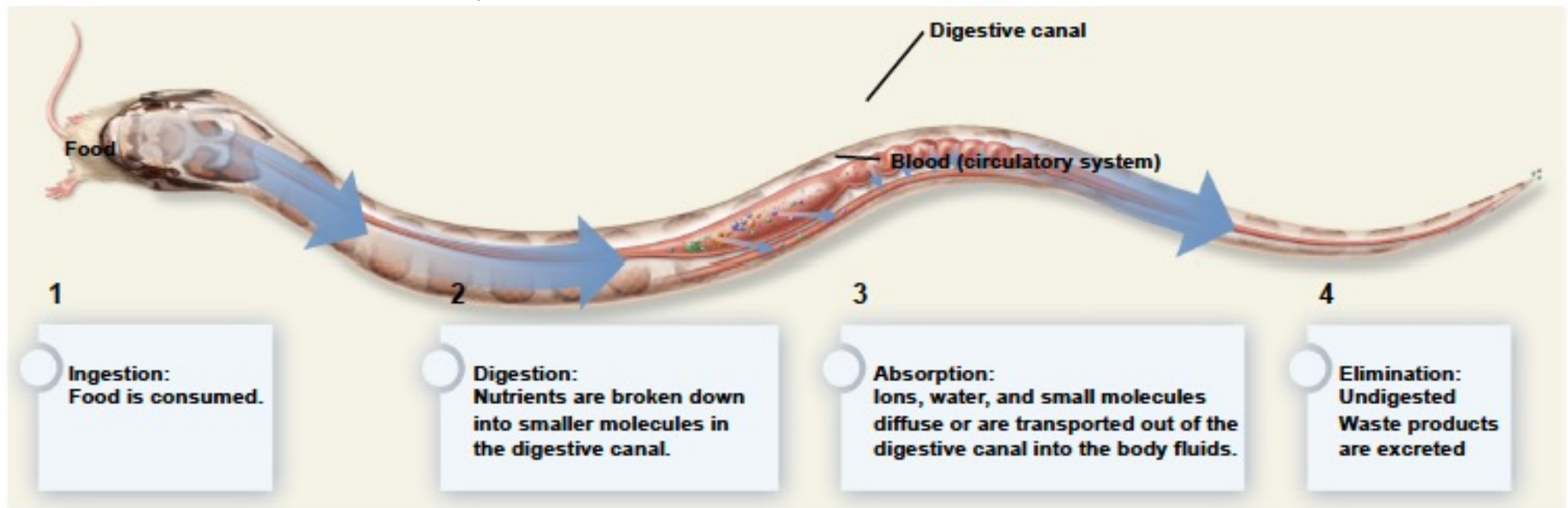


Agenda

- Nutrition
- Anatomical organization of gastro-intestinal function
- Mechanisms of nutrient's digestion and absorption
- Endocrine and neural control of gastro-intestinal function

Nutrition

- **Nutrition** - process of consuming and using food and nutrients
- **Nutrient** – any substance that is needed for survival, growth, development, tissue repair, or reproduction
- All organisms require nutrients to survive
- **Plants are autotrophs:** Make their own food
- All **animals are heterotrophs:** Must consume **food from other sources**
- 3 dietary categories: **Herbivore, carnivore, omnivore**
- Nutrition is in 4 steps:



Nutrizione Animale

- Animals require:
 - 5 categories of **organic nutrients**
 - Carbohydrates → glucose → carbon + energy
 - Proteins → amino acids → protein synthesis or energy
 - Lipids → membrane components, thermic insulation or energy
 - Nucleic acids → synthesis of DNA, RNA and ATP
 - Vitamins → coenzymes

Tabella 41.1 Principali nutrienti degli animali, risorse alimentari, alcune loro funzioni e sintomi dovuti alla loro carenza nei vertebrati			
Classe di nutrienti	Risorse alimentari	Funzioni nei vertebrati	Sintomi dovuti alla carenza
Carboidrati	Tutte le risorse alimentari, specialmente piante ricche di amido	Fonti di energia; componenti di alcune proteine; fonti di carbonio	Indebolimento muscolare; perdita di peso; formazione di chetoni (diminuzione del valore di pH nel sangue)
Proteine	Tutte le risorse alimentari, specialmente carne, legumi, cereali e radici	Forniscono gli amminoacidi per la sintesi di nuove proteine; alcuni amminoacidi sono utilizzati come fonte di energia	Perdita di peso; atrofia muscolare; debolezza; depressione del sistema immunitario; maggiore suscettibilità alle infezioni
Lipidi	Tutte le risorse alimentari, specialmente carni grasse, latticini e piante oleose	Principali componenti delle membrane; fonti di energia; isolanti termici; costituiscono i componenti di alcuni ormoni	Perdita dei capelli; perdita di peso; disturbi ormonali e riproduttivi
Acidi nucleici	Tutte le risorse alimentari	Forniscono gli zuccheri, le basi azotate e i fosfati utilizzati per la sintesi di DNA, RNA e ATP	Nessuno; i componenti degli acidi nucleici possono essere sintetizzati dalle cellule partendo da amminoacidi e zuccheri

- Inorganic nutrients**

- Water
- Minerals

- 4 groups of **essential nutrients** (cannot be synthesized from precursor molecules)
 - Essential amino acids
 - Essential fatty acids
 - Vitamins
 - Minerals

Essential amino acids

Essential Amino acids

- **8**(of total 20) **amminoacids** required by many animals:
 - Isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine
- **Cannot be synthesized** by animal's cells
- Are **not stored**
- Carnivores and omnivores readily **obtain all 8 in meat**
- **Herbivores have evolved essential amino acid synthesis ability**
- **Most plants do not contain** every essential amino acid in sufficient quantity
-> **Integration** is required in humans

Plant-based burgers



The **Impossible Burger** contains:

Water, **Soy Protein Concentrate**, **Coconut Oil**, **Sunflower Oil**, Natural Flavors, 2% or less of: Potato Protein, Methylcellulose, Yeast Extract, Cultured Dextrose, Food Starch Modified, **Soy Leghemoglobin**, Salt, Soy Protein Isolate, Mixed Tocopherols (Vitamin E), Zinc Gluconate, Thiamine Hydrochloride (Vitamin B1), Sodium Ascorbate (Vitamin C), Niacin, Pyridoxine Hydrochloride (Vitamin B6), Riboflavin (Vitamin B2), Vitamin B12.

The **Beyond Burger** contains:

Water, **Pea Protein Isolate**, Expeller-Pressed **Canola Oil**, **Refined Coconut Oil**, Contains 2% or less of the following: Cellulose from Bamboo, Methylcellulose, Potato Starch, Natural Flavor, Maltodextrin, Yeast Extract, Salt, Sunflower Oil, Vegetable Glycerin, Dried Yeast, Gum Arabic, Citrus Extract (to protect quality), Ascorbic Acid (to maintain color), **Beet Juice Extract (for color)**, Acetic Acid, Succinic Acid, Modified Food Starch, Annatto (for color).

Non esattamente come carne..

European Journal of Nutrition (2022) 61:461–475
<https://doi.org/10.1007/s00394-021-02658-1>

ORIGINAL CONTRIBUTION



Digestible indispensable amino acid score (DIAAS) is greater in animal-based burgers than in plant-based burgers if determined in pigs

Natalia S. Fanelli¹ · Hannah M. Bailey² · Tyler W. Thompson³ · Robert Delmore³ · Mahesh Narayanan Nair³ · Hans H. Stein^{1,2} 

Qualita' delle proteine (DIAAS):

Animal-based burger > Impossible burger (soia) > Beyond burger (piselli)

Integrazione necessaria per dieta con combinazione Impossible Burger + panino:

Bambini dai **6 mesi ai 3 anni**: lisina, leucina, treonina e metionina erano insufficienti

Individui **> 3 anni**: solo la lisina e' insufficiente.

Essential fatty acids

- **Certain polyunsaturated fatty acids** cannot be synthesized by animal cells
- **Found primarily in plants**
- **Strict carnivores obtain them from fish or adipose tissue** of birds and mammals
- **Linoleic acid** is a vital **precursor to arachidonic acid**
(present in the **phospholipids of membranes** of the cells and precursor of **prostaglandins involved in coagulation**)

Table 2 Sources of omega fatty acids

Omega-6

Linoleic acid Leafy vegetables, seeds, nuts, grains, vegetable oils (corn, safflower, soy bean, cottonseed, sesame, sunflower).

Omega-3

Linolenic acid Fats and oils (canola, soy bean, walnut, wheat germ).
Nuts and seeds (butternuts, walnuts, soy bean kernels).
Vegetables (soy beans).

EPA and DHA Human milk
Shellfish and fish (mackerel, tuna, salmon, bluefish, mullet, sturgeon, herring, trout, sardines)
(or can be made out of linolenic acid).

(Source: Whitney&Rolfes,1996)

EPA: Eicosapentanonic Acid; DHA: Docosahexaeonic acid

Vitamins

- **Vitamins** are important **organic nutrients** that serve as **coenzymes**
 - **Water-soluble** vitamins:
 - vitamin C - **not stored**
 - **Fat-soluble** vitamins:
 - vitamin A – **stored in adipose tissue**
- Not all animals require the same vitamins: → **Between vertebrates only primates and guinea pigs can't synthesize vitamin C**



Tabella 41.2 Vitamine indispensabili per gli animali			
Classe di nutrienti	Risorse alimentari	Funzioni nei vertebrati	Sintomi dovuti alla carenza
Vitamine idrosolubili			
Biotina	Fegato; legumi; semi di soia; uova; noci; funghi; alcune verdure verdi	Coenzima necessario per la gluconeogenesi e per la sintesi degli acidi grassi; necessaria per catabolizzare alcuni aminoacidi	Prurito cutaneo; nausea; perdita dell'appetito; disturbi mentali (depressione o allucinazioni)
Acido folico	Verdure verdi; noci; legumi; grano in chicchi; frattaglie (specialmente fegato; reni; cuore)	Coenzima necessario per la sintesi degli acidi nucleici	Anemia; depressione; difetti congeniti
Niacina	Legumi; noci; latte; uova; carne	Partecipa in molte reazioni di ossidoriduzione	Prurito cutaneo; diarrea; confusione mentale; perdita della memoria
Acido pantotenico	Quasi in tutti i cibi	Parte del coenzima A, che partecipa a numerose reazioni di sintesi, inclusa la formazione di colesterolo	Sensazione di bruciore ai piedi e alle mani; sintomi gastrointestinali; depressione
Vitamina B ₁ (tiamina)	Carne; legumi; grano in chicchi	Necessaria per alcuni passaggi del metabolismo intermedio, genera la produzione di energia da alcuni aminoacidi	Beriberi (indebolimento muscolare, anemia, problemi cardiaci, perdita di peso)
Vitamina B ₂ (riboflavina)	Latticini; carne; frattaglie; cereali; alcune verdure	Coenzima respiratorio; necessaria per il metabolismo dei grassi, dei carboidrati e delle proteine	Seborrea (eccessiva secrezione oleosa delle ghiandole cutanee, che provoca lesioni della pelle)
Vitamina B ₆ (piridossina)	Carne; fegato; pesce; noci; grano in chicchi; legumi	Cofattore per almeno 100 enzimi coinvolti nel metabolismo degli aminoacidi, nel metabolismo dei lipidi e nella sintesi dell'eme	Seborrea; malattie di origine nervosa; depressione; confusione mentale; spasmi muscolari
Vitamina B ₁₂	Carne; fegato; uova; alcuni molluschi e crostacei; latticini	Necessaria per la formazione dei globuli rossi	Anemia; disordini del sistema nervoso che comportano problemi sensoriali; problemi nell'andatura e nell'equilibrio; perdita del controllo della vescica e dell'intestino
Vitamina C (acido ascorbico)	Agrumi; verdure verdi; pomodori; patate	Antiossidante e capace di neutralizzare i radicali liberi; facilita l'assorbimento del ferro; aiuta a mantenere sani il tessuto connettivo e le gengive	Scorbuto (malattie del tessuto connettivo associate con lesioni cutanee, debolezza, scarsa rimarginazione delle ferite, caduta dei denti); sanguinamento delle gengive
Vitamine liposolubili			
Vitamina A (retinolo)	Fegato; verdure verdi e gialle; alcuni frutti in piccole quantità	Componente del pigmento visivo; regolatore della trascrizione; importante nella riproduzione e nell'apparato immunitario	Cecità notturna dovuta alla perdita della capacità visiva; lesioni cutanee; depressione immunitaria
Vitamina D	Olio di pesce; pesce; tuorlo d'uovo; fegato; sintetizzata nella cute attraverso il sole	Necessaria per l'assorbimento di calcio e fosforo nell'intestino; crescita delle ossa	Rachitismo (ossa deboli e deformate) nei bambini; osteomalacia (ossa deboli) negli adulti
Vitamina E	Carne; oli vegetali; cereali; alcuni frutti e verdure in piccole quantità	Antiossidante; inibisce la sintesi delle prostaglandine	Sconosciuti, possibile atrofia muscolo-scheletrica; disordini del sistema nervoso periferico
Vitamina K	Legumi; verdure verdi; alcuni tipi di frutta; alcuni oli vegetali (olio di oliva; olio di semi di soia); fegato; anche sintetizzata dai batteri presenti nella parte terminale dell'intestino	Componente del processo della coagulazione sanguigna	Ridotta capacità di coagulazione

Minerals

- Minerals are inorganic ions
 - Many required in only trace amounts (<1 mg/day)
 - Some minerals can be stored (Iodine → thyroid; and calcium → bones or shells)
 - Not all minerals used at the same rate



Tabella 41.3 Minerali indispensabili per gli animali			
Classe di nutrienti	Risorse alimentari	Funzioni nei vertebrati	Sintomi dovuti alla carenza
<u>Calcio (Ca)</u>	Latticini; cereali; legumi; grano in chicchi; verdure verdi a foglia; ossa (mangiate da alcuni animali)	Formazione delle ossa e dei denti; esocitosi di secrezioni accumulate nei nervi e in altre cellule; contrazione muscolare; coagulazione del sangue	Malattie muscolari; diminuzione della massa ossea; crescita ridotta nei bambini
<u>Cloro (Cl)</u>	Carne; latticini; sangue; saline	Partecipa al mantenimento dell'equilibrio elettrico, acido-base e osmotico attraverso le membrane, soprattutto nel cuore e nelle cellule del sistema nervoso	Malattie del sistema muscolare e di quello nervoso
<u>Cromo (Cr)</u>	Fegato; frutti di mare; alcune noci; carne; funghi; alcune verdure	Necessario per un corretto metabolismo del glucosio, forse aiutando l'azione dell'ormone insulina	Alterazioni nell'equilibrio dei lipidi e del glucosio nel sangue
<u>Rame (Cu)</u>	Pesce; molluschi; noci; legumi; fegato e frattaglie di altri organi	Necessario per la produzione di emoglobina e per la sintesi della melanina; necessario per la formazione del tessuto connettivo; serve come componente che lega l'ossigeno in alcuni invertebrati	Anemia (anomalo basso numero di globuli rossi); modificazioni delle ossa
<u>Iodio (I)</u>	Alghe marine; molluschi; latte; sale contenente iodio	Necessario per la formazione degli ormoni tiroidei	Incapacità di sintetizzare gli ormoni tiroidei, con conseguente aumento del volume della tiroide
<u>Ferro (Fe)</u>	Fegato e frattaglie di altri organi; alcuni tipi di carne; uova; legumi; verdure verdi a foglia	Componente capace di legare l'ossigeno nell'emoglobina; cofattore in alcuni enzimi	Anemia
<u>Magnesio (Mg)</u>	Fieno; erba; grano in chicchi; verdure verdi a foglia	Cofattore in molti enzimi che utilizzano l'ATP come substrato	Modificazioni delle funzioni del sistema nervoso
<u>Manganese (Mn) e molibdeno (Mo)</u>	Noci; grano in chicchi; legumi; verdure; fegato	Cofattori per molti enzimi	Scarsa crescita; formazione anomala dello scheletro; malattie del sistema nervoso (convulsioni)
<u>Fosforo (P)</u>	Latticini; grano; legumi; noci; carne	Formazione delle ossa e dei denti; componente di DNA, RNA e ATP	Osteoporosi; indebolimento muscolare
<u>Potassio (K)</u>	Carne; frutta; verdure; latticini; grano	Le stesse del cloro	Indebolimento muscolare; gravi aritmie cardiache; sintomi gastrointestinali
<u>Selenio (Se)</u>	Frutti di mare; uova; pollo; semi di soia; grano	Antiossidante; cofattore per molti enzimi	Malattia di Keshan (danneggiamento e perdita della muscolatura cardiaca)
<u>Zolfo (S)</u>	Tutti i tipi di proteine	Componente di due amminoacidi (metionina e cisteina)	Incapacità di sintetizzare molte proteine
<u>Sodio (Na)</u>	Molti tipi di frutta; verdura; carne; saline	Le stesse del cloro	Crampi muscolari; modificazioni dell'attività dei nervi
<u>Zinco (Zn)</u>	Presente in grande quantità nella carne; pesce; molluschi (ostriche); grano	Molte funzioni collegate alla riparazione dei tessuti; produzione dello sperma; cofattore per molti enzimi metabolici; consente a certi fattori di trascrizione di legarsi al DNA	Nanismo; perdita di alcune sensazioni come il gusto; depressione immunitaria; lesioni cutanee

Principles of digestion

- **Intracellular digestion**
 - Only in **sponges** and **some single-celled organisms**
 - **Uses phagocytosis** to bring food particles directly into a cell
 - No mechanism for storing food
- **Extracellular digestion**
 - **Protects interior of cells** from hydrolytic enzymes
 - Can consume large food
 - Generally occurs **in a cavity**
- **Gastro-vascular cavity**
 - **Both Digestion** (gastro) and **Distribution** (vascular) of nutrients
 - Simple extracellular digestion typically in **Hydra**
 - **One opening** is entry and exit
 - Food particles/nutrients eventually phagocytosed

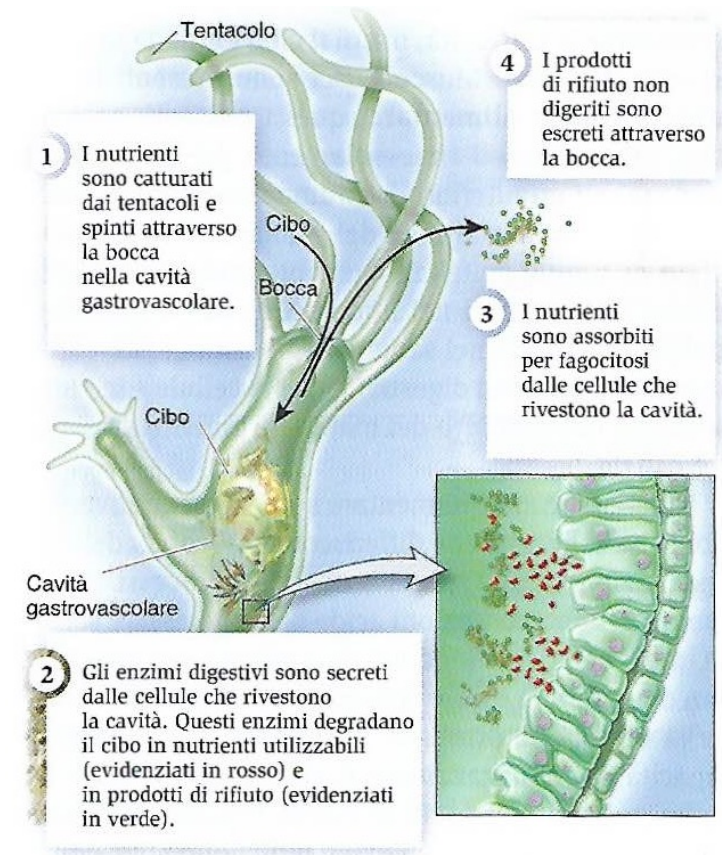
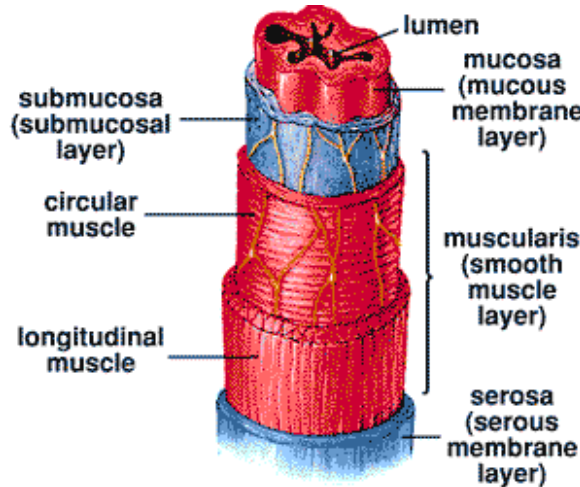


Figura 41.5 La digestione extracellulare in una cavità gastrovascolare. Negli animali provvisti di cavità gastrovascolare, come l'*Hydra*, lo cnidario qui raffigurato, la digestione avviene all'interno della cavità, a livello extracellulare.

Alimentary canal in vertebrates

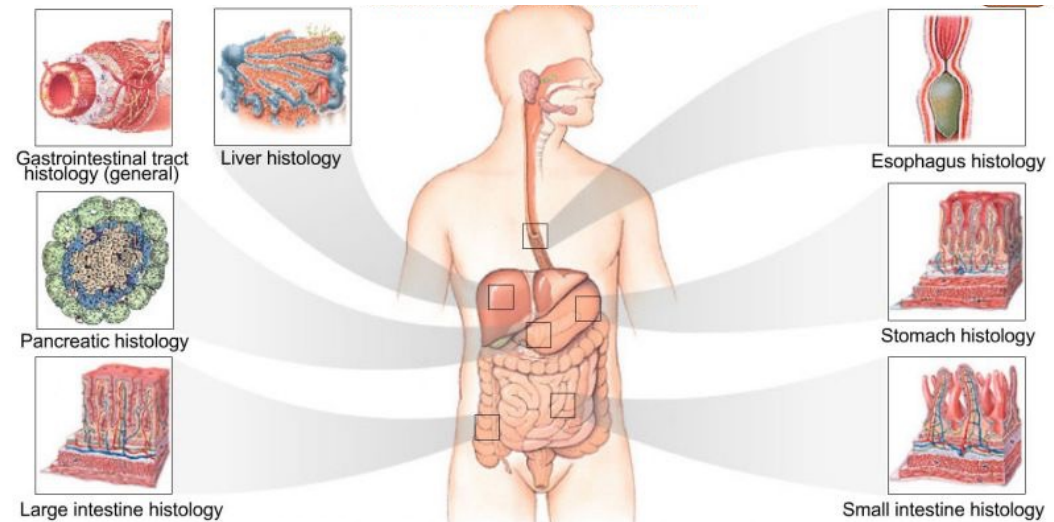
- **Single elongated tube** with distinct **entry and exit** ends, lined by **epithelial cells** that

- Transport of digested material
- Synthesize and secrete digestive enzymes
- Secrete hormones



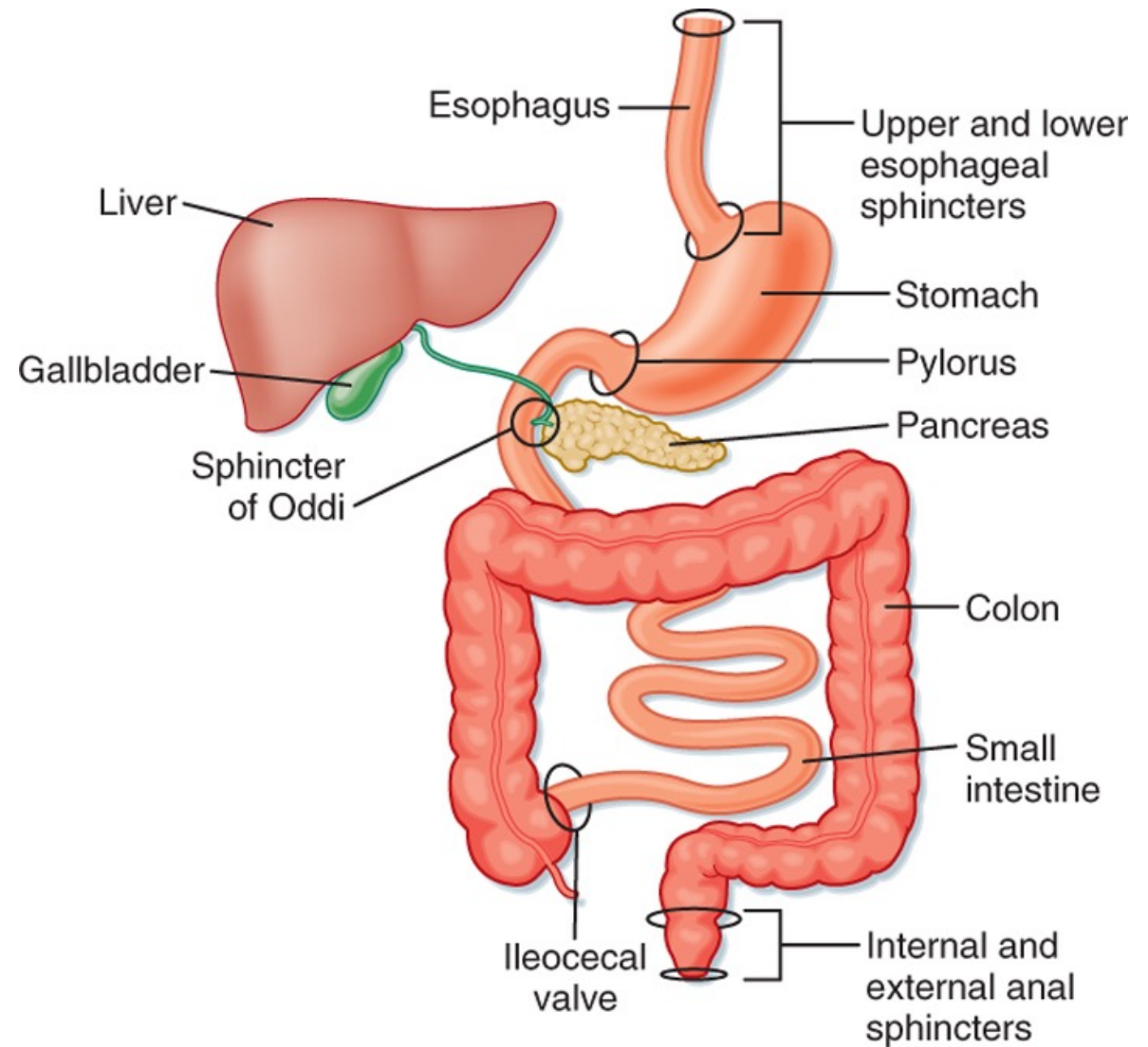
- **Hollow tube** composed of a **lumen** whose diameter varies.
- Surrounded by a wall made up of 4 principal layers:
 - **Mucosa**
 - Epithelial lining; A lamina propria of loose connective tissues rich in blood, lymph vessels and smooth muscle cells; Muscularis mucosae.
 - **Submucosa**
 - Dense connective tissues with many blood and lymph vessels.
 - **Muscularis**
 - Contains smooth muscle cells, divide into 2 layers; internal (circular); external (longitudinal)
 - **Serosa**
 - Thin layer of loose connective tissue rich in blood and lymph vessels and adipose and single squamous covering epithelium (mesothelium)

- Several **specialized regions**
→ Different environments for different processes and Storage area



Functional regions

- **Anterior** end functions primarily in **ingestion**
 - Oral cavity, salivary glands, pharynx (throat), and esophagus
- **Middle** portion functions in storage and **initial Digestion**:
 - Stomach
 - Upper part of small intestine and associated organs (liver, gallbladder, pancreas)
- **Posterior** part functions in **final digestion, absorption, and elimination**
 - Remainder of small intestine and large intestine



Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
Copyright © 2008 by Mosby, an imprint of Elsevier, Inc. All rights reserved



Mouth

- Processing may be extensive (herbivores) or not (carnivores)
- Saliva begins initial processing of food
 - Moisten and lubricate food to facilitate swallowing
 - Dissolve food particles to facilitate taste
 - Kill ingested bacteria
 - Initiate digestion of polysaccharides with amylase (not very important)

Pharynx and esophagus

Pathway to stomach only

Do not contribute to digestion or absorption

Swallowing begins in esophagus as voluntary action

Action continues by involuntary peristalsis



Stomach

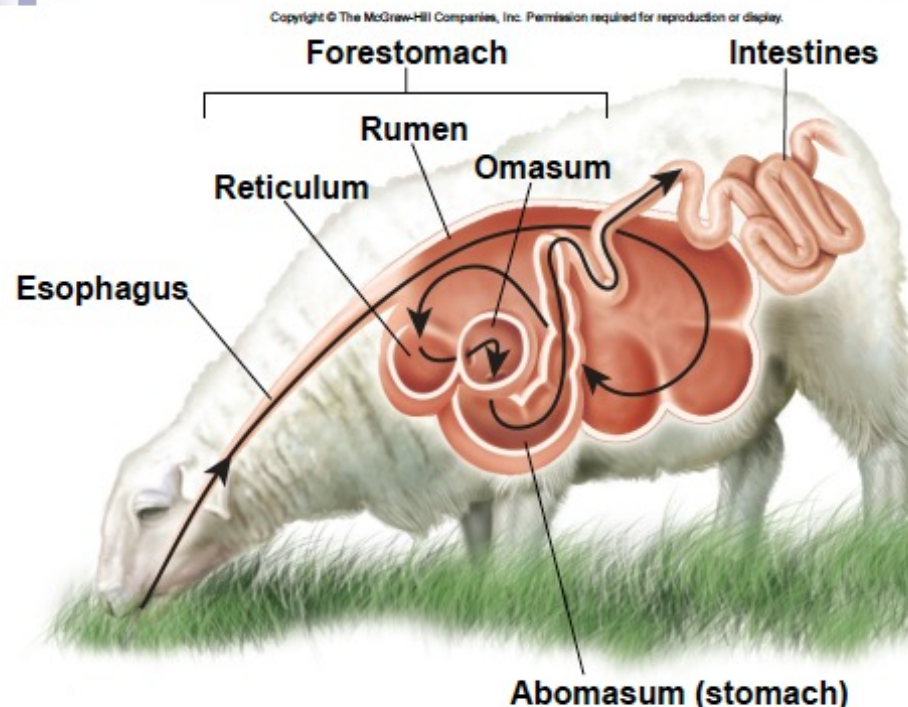
- Saclike organ evolved for storing food
- Muscular nature helps break up food
- Partially digests proteins
- Regulates rate of emptying into small intestine
- Glands secrete
 - Hydrochloric acid – kills microbes, dissolves particulate matter
 - Pepsinogen – converted to pepsin to begin protein digestion
- No lipid or carbohydrate digestion
- Food reduced to chyme

HCL secretion in the stomach



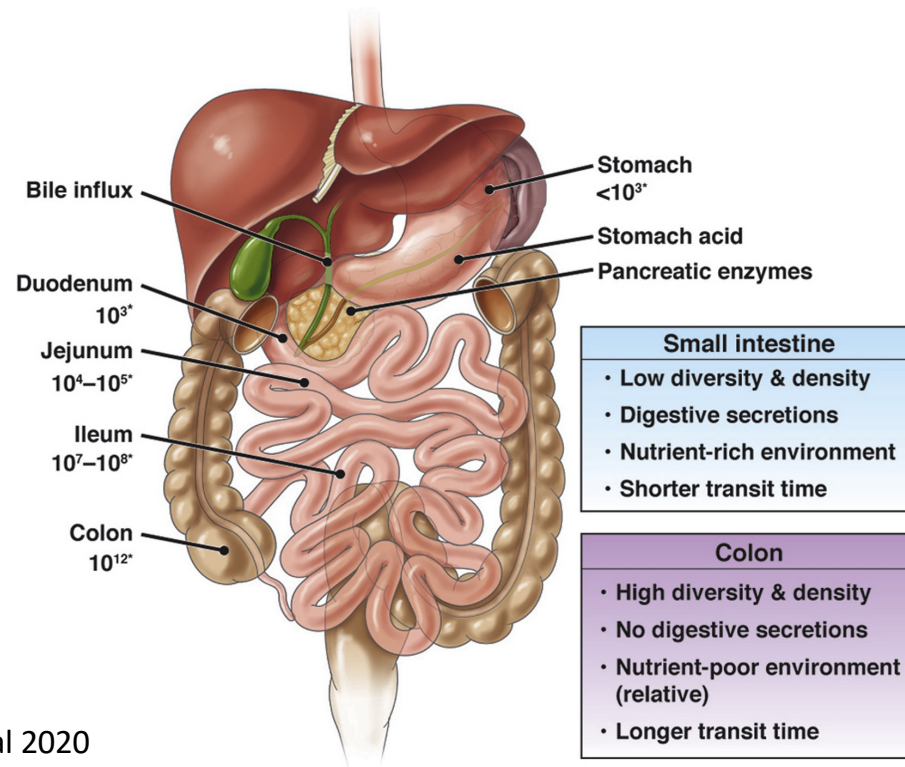
Herbivore stomachs

- **Herbivores** must **digest cellulose** into monosaccharides but **lack cellulase**
- Rely on **microbes** to digest cellulose
- In many herbivores such as **cow**:
 - **Fore-stomach** – 3 lower esophageal pouches:
 - **Rumen** and **reticulum** → contain **cellulose digesting microbes**
 - **Omasum** → **absorbs** some of the water and salts from food
 - **Abomasum** – **true stomach** →> Contains **acid and proteolytic enzymes** to process eventually food, microbes and by-products of microbial digestion



Small intestine

- **Digestion and absorption of food and water**
- **Hydrolytic enzymes** on luminal surface or secreted by pancreas into lumen
- Products of digestion **absorbed across epithelial cells and enter blood**
- Vitamins, mineral and water also absorbed
- Contain **bacteria** (but lower density than large intestine)



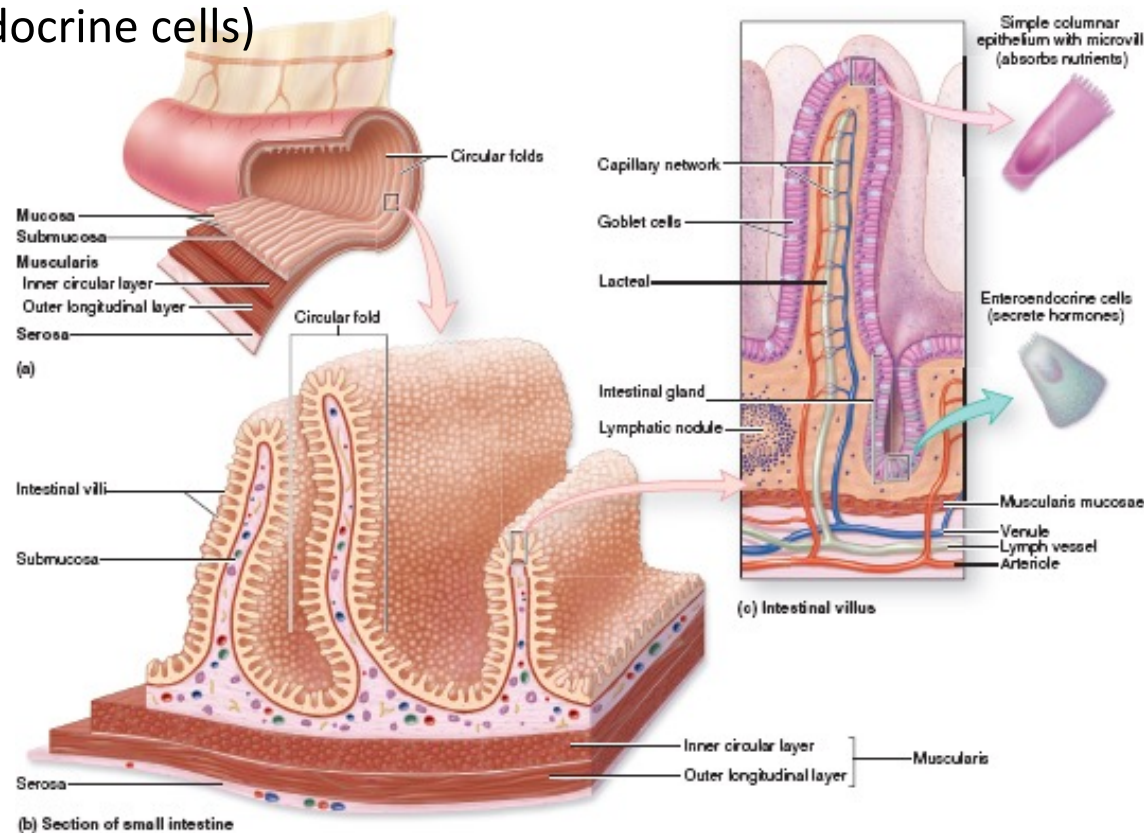
Kastl et al 2020

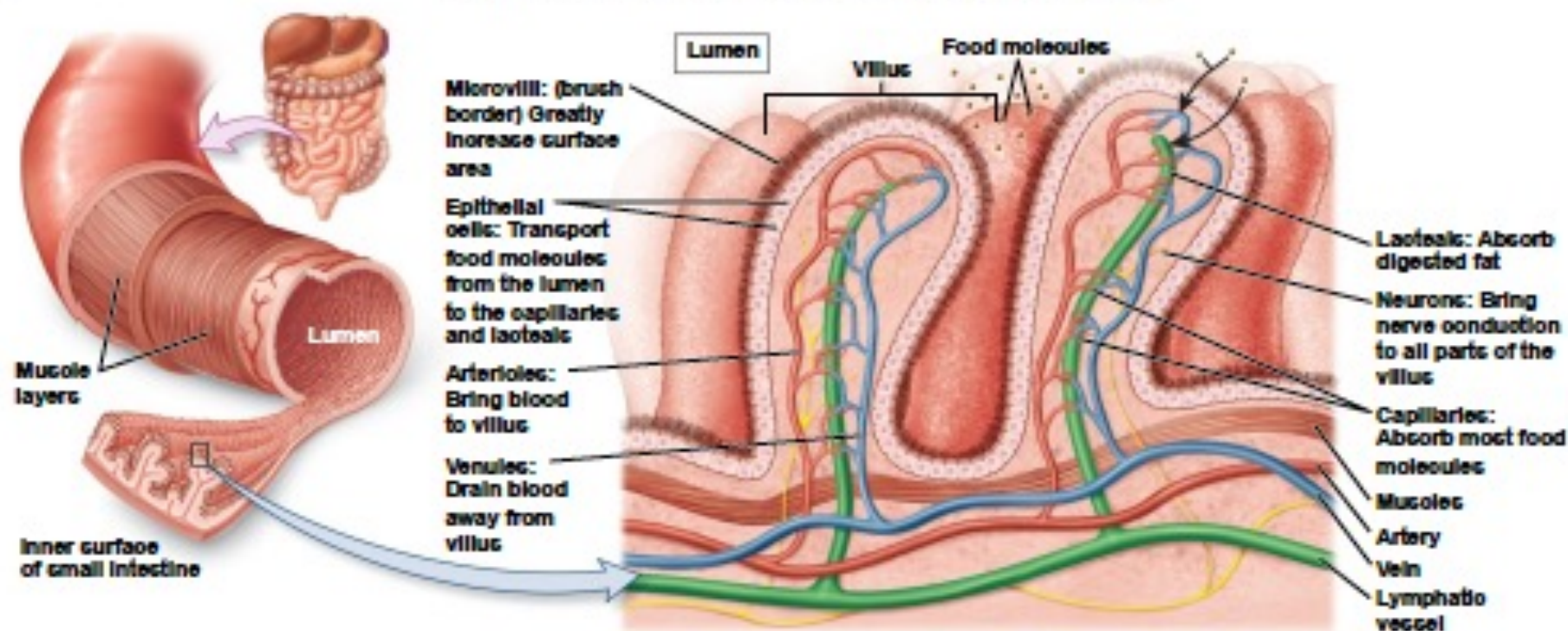
Figure 1. Overview of bacterial colony density throughout the digestive tract, factors from the proximal tract that shape small intestinal communities, and distinguishing characteristics of the small intestine and colon.

Small intestine

Specialized for increased surface area (~600-fold)

- Mucosa is folded → **Plicae**
- Plicae present **finger-like projections** → **Villi**
- **Epithelial cells covered with microvilli** → brush border
- Contains also other specialized cells involved in secretion (enteroendocrine cells)



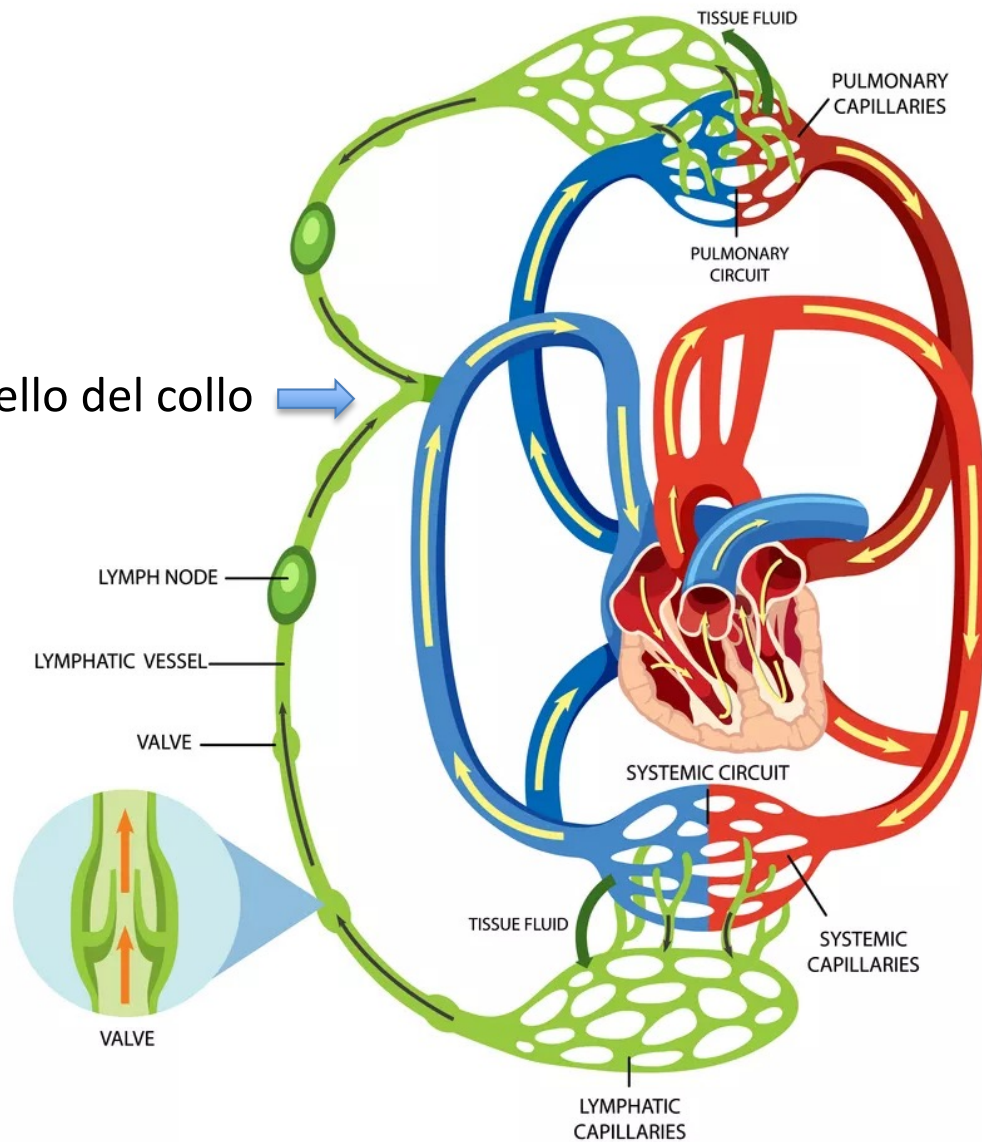


- Each villus has a
 - Capillary – nutrients other than fat absorbed into blood
 - Lacteal (lymphatic vessel) – allows for larger fat particles to enter, eventually dumped into blood

LYMPHATIC CIRCULATION

Rimette nelle vene cave a livello del collo →

Valvole garantiscono unidirezionalità' →

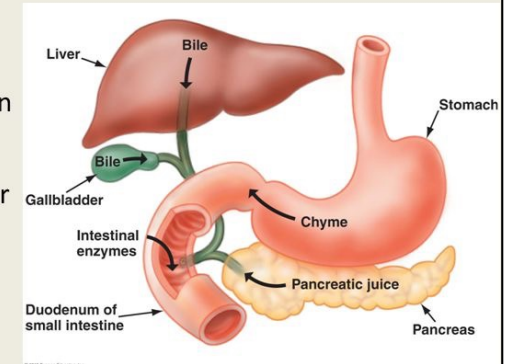


Small intestine and Accessory organs

ORGAN	MAJOR FUNCTIONS*	COMMENTS/ADDITIONAL FUNCTIONS
Mouth and associated accessory organs	Ingestion: food is voluntarily placed into oral cavity Propulsion: voluntary (buccal) phase of deglutition (swallowing) initiated by tongue; propels food into pharynx Mechanical breakdown: mastication (chewing) by teeth and mixing movements by tongue Digestion: salivary amylase in saliva, produced by salivary glands, begins chemical breakdown of starch	Mouth serves as a receptacle; most functions performed by associated accessory organs. Mucus in saliva helps dissolve foods so they can be tasted and moistens food so that tongue can compact it into a bolus that can be swallowed. Saliva cleanses and lubricates oral cavity and teeth.
Pharynx and esophagus	Propulsion: peristaltic waves move food bolus to stomach, thus accomplishing involuntary (pharyngeal-esophageal) phase of deglutition	Primarily food chutes; mucus produced helps to lubricate food passageways.
Stomach	Mechanical breakdown and propulsion: peristaltic waves mix food with gastric juice and propel it into the duodenum Digestion: pepsin begins the digestion of proteins Absorption: absorbs a few fat-soluble substances (aspirin, alcohol, some drugs)	Also stores food until it can be moved into the duodenum. Hydrochloric acid produced is a bacteriostatic agent and activates protein-digesting enzymes. Mucus produced helps lubricate and protect stomach from self-digestion. Intrinsic factor produced is required for intestinal absorption of vitamin B ₁₂ .
Small intestine and associated accessory organs (liver, gallbladder, pancreas)	Mechanical breakdown and propulsion: segmentation by smooth muscle of the small intestine continually mixes contents with digestive juices and, along with short-distance peristaltic waves, moves food along tract, allowing sufficient time for digestion and absorption Digestion: digestive enzymes delivered from pancreas and brush border enzymes attached to microvilli membranes complete digestion of all classes of foods Absorption: breakdown products of carbohydrate, protein, fat, and nucleic acid digestion, plus vitamins, electrolytes, and water, are absorbed by active and passive mechanisms	Small intestine is highly modified for digestion and absorption (circular folds, villi, and microvilli). Alkaline mucus produced by intestinal glands and bicarbonate-rich juice ducted in from pancreas help neutralize acidic chyme and provide proper environment for enzymatic activity. Bile produced by liver emulsifies fats and enhances (1) fat digestion and (2) absorption of fatty acids, monoglycerides, cholesterol, phospholipids, and fat-soluble vitamins. Gallbladder stores and concentrates bile, releasing it to small intestine in response to hormonal signals.
Large intestine (cecum, colon, rectum)	Digestion: some remaining food residues are digested by enteric bacteria (which also produce vitamin K and some B vitamins) Absorption: absorbs most remaining water, electrolytes (largely NaCl), and vitamins produced by bacteria Propulsion: propels feces toward rectum by haustral churning and mass movements Defecation: reflex triggered by rectal distension; eliminates feces from body	Temporarily stores and concentrates residues until defecation can occur. Copious mucus produced by goblet cells eases passage of feces through colon.

Accessory organs

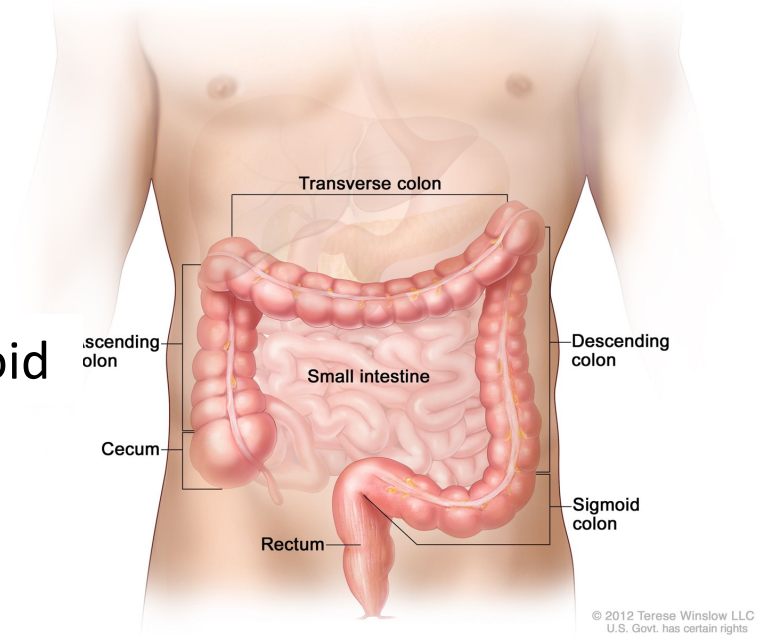
- Role of liver
 - Produces bile to assist in fat digestion
- Role of gall bladder
 - Stores bile from liver
- Role of pancreas
 - Produces bicarbonate
 - Produces digestive enzymes



Pancreatic Juice (from pancreas)			Bile (from liver)
Proteases (digests proteins)	Amylase (digests carbohydrates)	Lipase (digests fats)	Bile Salts (digests fats)
Trypsin Chymotrypsin	Carboxypeptidase		
↓	↓		
Proteins → Peptides → Amino acids	Starch → Maltose	Dietary fat Triglycerides	Monoglycerides Fatty acids

Large intestines

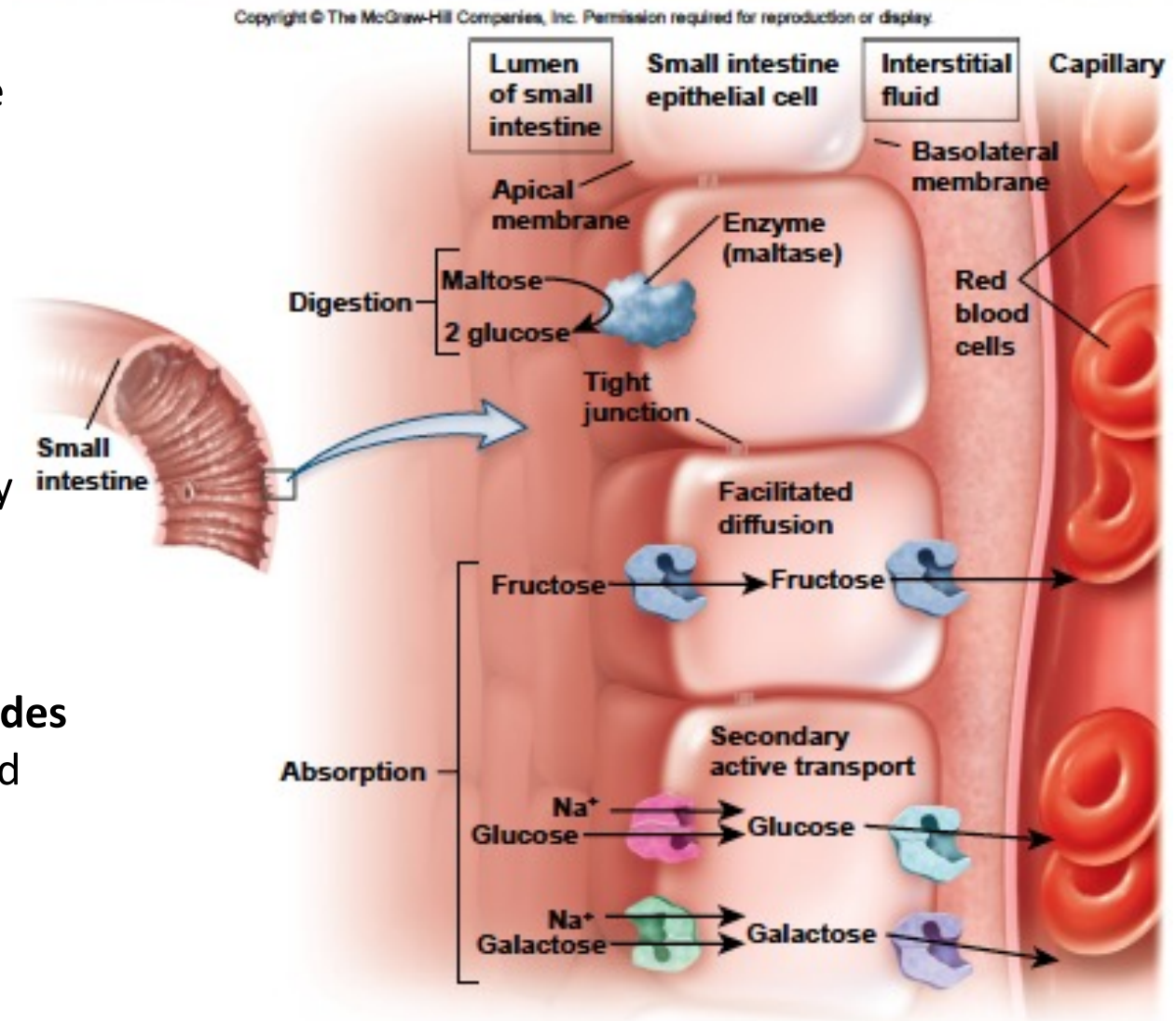
- Primary function to store and concentrate fecal matter and absorb some salt and water
- Bacteria will produce vitamins and flatus
- Cecum
 - Chyme enters through sphincter
 - Appendix
- Colon
 - Ascending, transverse and descending, sigmoid
- Rectum
- Anus (internal/external sphincter)



Mechanisms of digestion and absorption

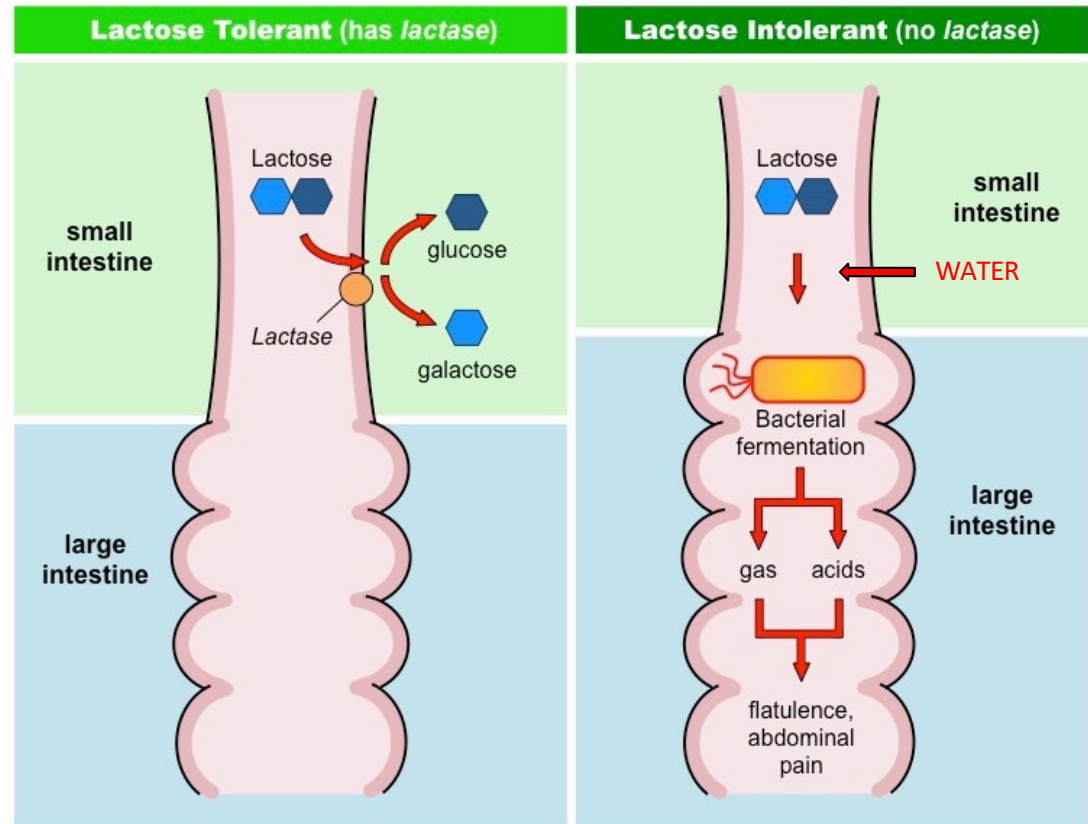
CARBOHYDRATES

- In typical omnivore, most are polysaccharides, starch and cellulose with some monosaccharides and disaccharides
- Mouth – **starch digestion** by **salivary amylase**
- Additional starch digestion by **pancreatic amylase**
- **Enzymes associated with epithelial cells** (for example maltase) **form monosaccharides**
- Monosaccharides transported **across intestinal epithelium into the blood**



Genetics Explains Lactose Intolerance

- Milk of mammals contains lactose
- **Lactase** digests lactose
- Once weaned, mammals never again drink milk, except for humans
- **90% of world's population can't fully digest lactose** after childhood due to **down-regulation of lactase**
- **Inability to digest lactose** promotes **water retention** and **bacterial fermentation** → **diarrhea, flatulence, pain**
- Developmental **mechanism** to turn off lactase **not established**
- Ability to digest lactose linked to genetic background → In Neolithic times (**farming** establishment), adults able to digest lactose enjoyed **selective advantage** and passed on their genes more frequently





Celiachia,
intolleranza e
allergia sono
tre disturbi
diversi, le cui
cause sono
ben distinte.



La **celiachia** è una malattia autoimmune su base infiammatoria dell'intestino tenue; colpisce solo i soggetti geneticamente predisposti che, di fronte all'assunzione di **glutine**, producono anticorpi che vanno a distruggere i delicati **villi**, strutture responsabili dell'assorbimento di nutrienti e minerali. Questo danno determina nei casi più gravi una **sindrome da malassorbimento** e uno stato di **malnutrizione**.

I **celiaci** pertanto sono obbligati a sospendere l'assunzione di **glutine** per tutta la vita: è importante che consumino prodotti con la dicitura "senza glutine", anche perché non tollerano contaminazioni. **Farine senza glutine** che possono essere consumate dai soggetti con celiachia sono per esempio: farina, crema e amido di riso, farina di mais, maizena, farina di miglio, farina di soia, di tapioca, di castagne, di ceci e altri legumi.

La malattia celiaca **può esordire a qualsiasi età**, anche negli anziani. Non sono tuttavia ancora noti i fattori che scatenano la malattia dopo anni di tolleranza al glutine.

Attualmente si calcola che la condizione celiaca interessi circa **l'1% della popolazione mondiale** e che più del 70% dei pazienti sia donna. Le stime ufficiali sulla prevalenza di questa malattia indicano che in Italia sono 182.858 i pazienti ai quali oggi è effettivamente diagnosticata la patologia, ma si constano circa 600.000 malati senza diagnosi. Questo dato risulta coerente con i riscontri epidemiologici provenienti da altre regioni del mondo. (dati tratti da Relazione annuale al Parlamento sulla celiachia 2016, Ministero della Salute).

Intolleranza

L'intolleranza non è mediata dall'immunità ed è un fenomeno di ipersensibilità. Si scatena, in un certo senso, un meccanismo di accumulo, per cui un **eccesso di glutine genera l'insorgenza di sintomi** quali diarrea, gonfiore addominale, mal di pancia, crampi, perdita di peso.

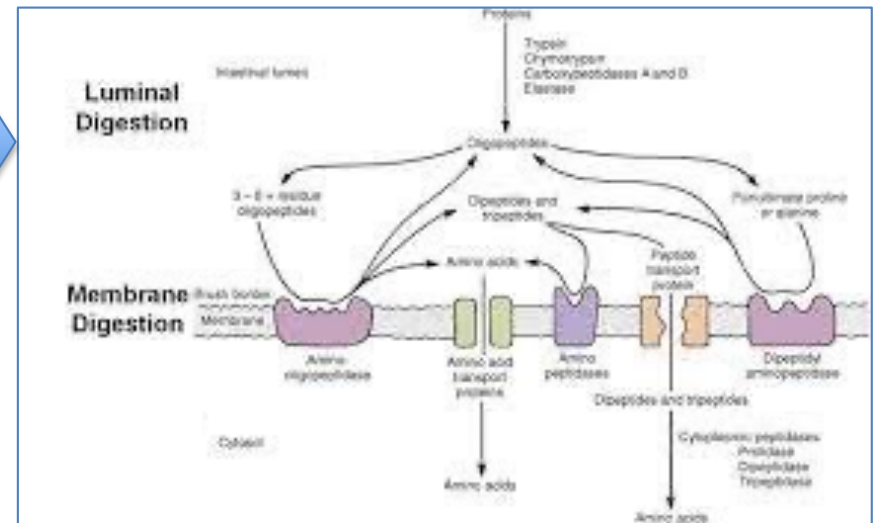
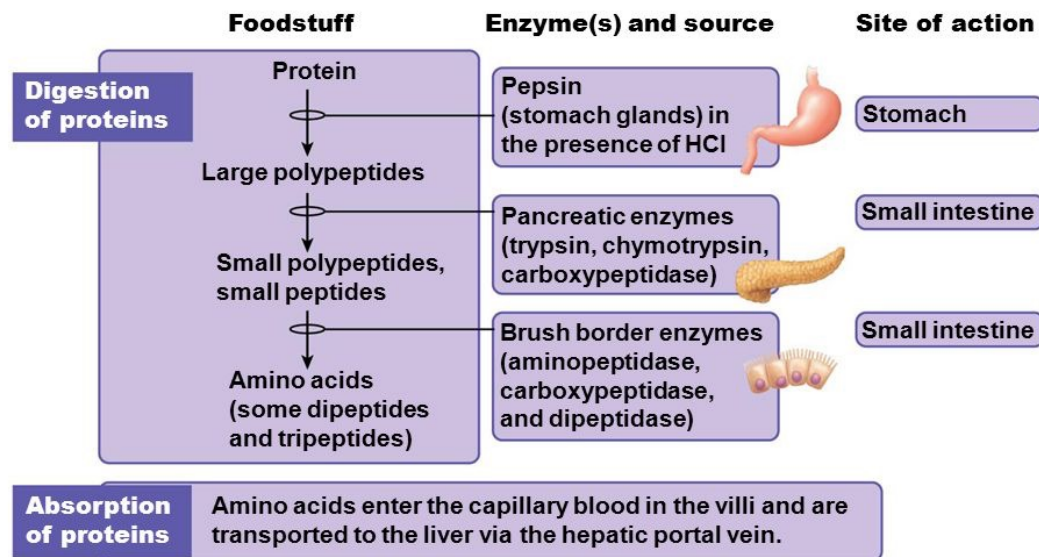
Allergia

L'allergia al glutine ha le caratteristiche di tutte le altre allergie alimentari e si presenta con **reazioni acute** di tipo anafilattico (orticaria, angioedema, asma, dolore addominale con o senza vomito e diarrea esplosiva, shock) in stretta correlazione temporale (minuti) con l'ingestione di glutine. È mediata dal **sistema immunitario**, in

Mechanisms of digestion and absorption

PROTEINS

- **Pepsin** in the **stomach**
- **Pancreas** secretes **trypsin** and **chymotrypsin**
- **Small intestine enzymes** complete **digestion to amino acids**
- Transported into intestinal cells and into blood stream



Mechanisms of digestion and absorption

- **Pepsin** in the stomach is **activated by H^+**
- **Trypsin, chymotrypsin** etc are **secreted as inactive precursors** and activated in small intestine in a cascade **process initiated by enteropeptidases** produced by parietal cells

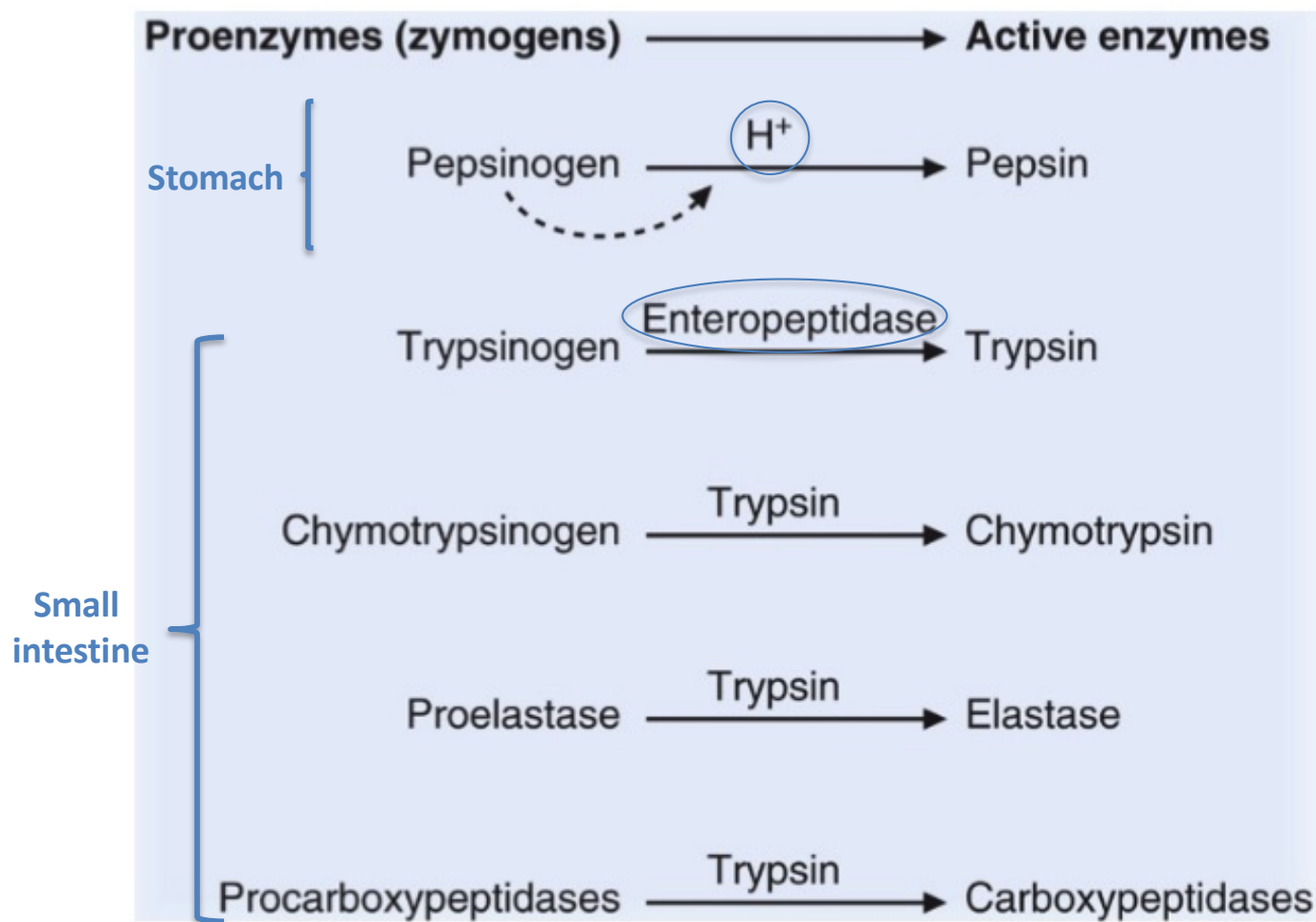
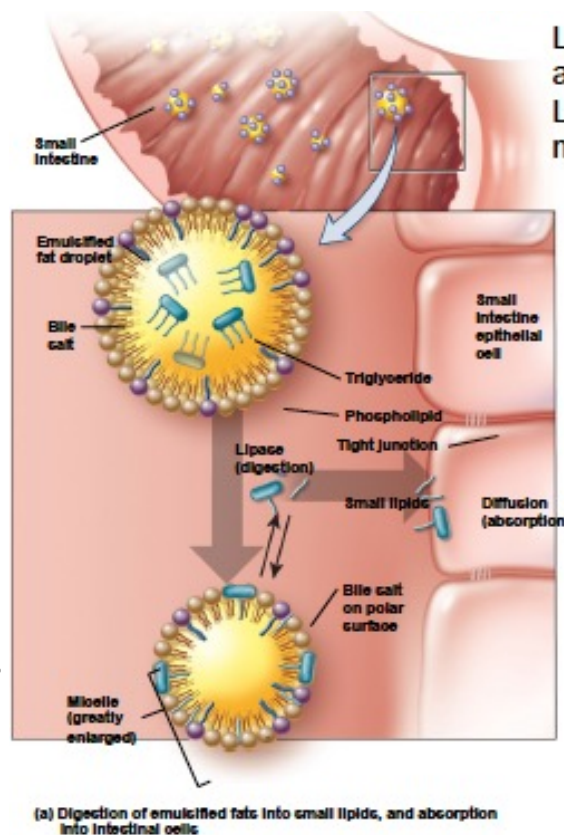


Fig.3. Digestion of Proteins by Different Enzymes

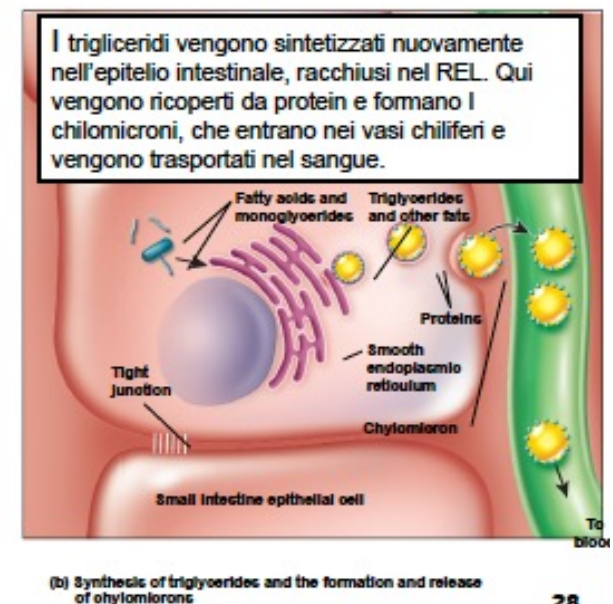
Mechanisms of digestion and absorption

LIPIDS

- Most in the form of triglycerides
- **Digestion entirely in small intestine (not in stomach)**
 - **Bile salts and GI movements emulsify fat into small droplets with increased surface area for digestion**
 - **Pancreatic lipase break triglycerides**
 - **Bile salts form small micelles** allowing lipids to diffuse into intestinal cells
- **Triglycerides reform inside intestinal cells and aggregate into chylomicrons in REL** and then released by **exocytosis** to pass **into lacteals** and then enter into **blood circulation**



La lipasi pancreatica digerisce i trigliceridi in 2 acidi grassi e un monogliceride
La bile ed i fosfolipidi aiutano a formare le micelle (4-7 nm)



I trigliceridi vengono sintetizzati nuovamente nell'epitelio intestinale, racchiusi nel REL. Qui vengono ricoperti da protein e formano i chilomicroni, che entrano nei vasi chiliferi e vengono trasportati nel sangue.

REL:reticolo endoplasmatico liscio

Mechanisms of digestion and absorption

VITAMINS, MINERALS

- Do **not require digestion**
- Absorbed in complete form
- **Fat-soluble** vitamins follow **pathway for fat absorption**



WATER

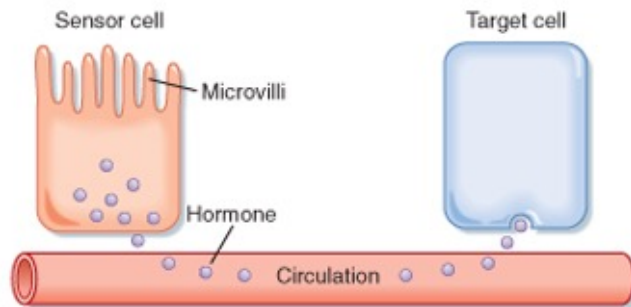
- Small amounts of water absorbed in stomach but **most in small intestine**



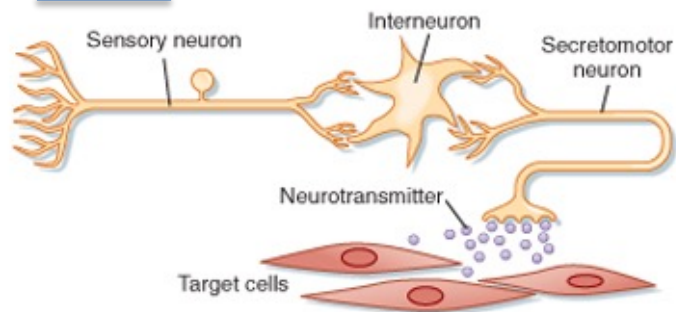
Control of GI tract

THREE MECHANISMS OF COMMUNICATION
MEDIATE RESPONSES IN THE GI TRACT

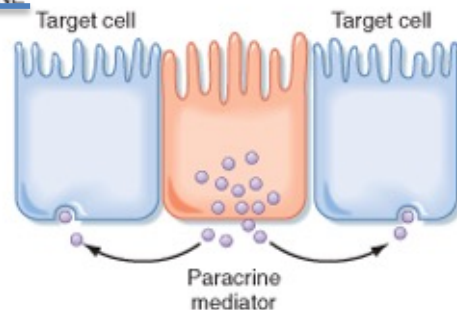
ENDOCRINE



NEUROCRINE

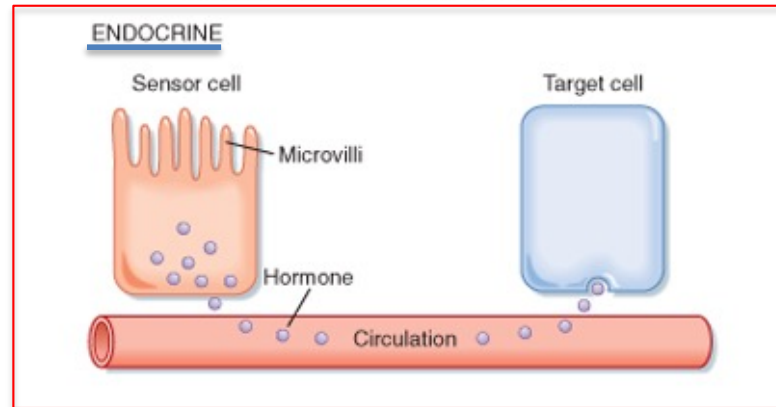


PARACRINE

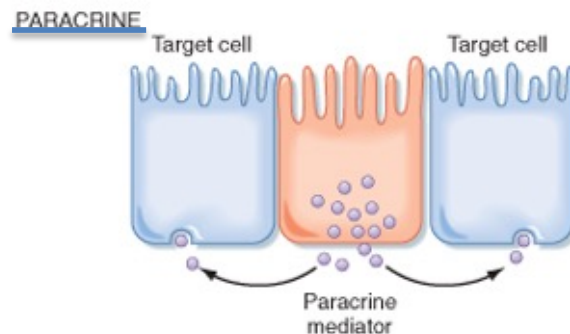
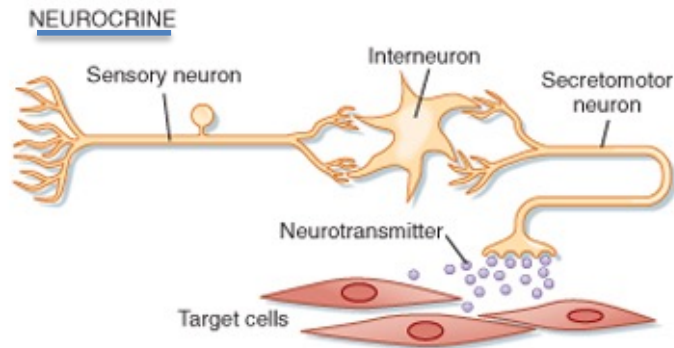


Control of GI tract

THREE MECHANISMS OF COMMUNICATION MEDIATE RESPONSES IN THE GI TRACT

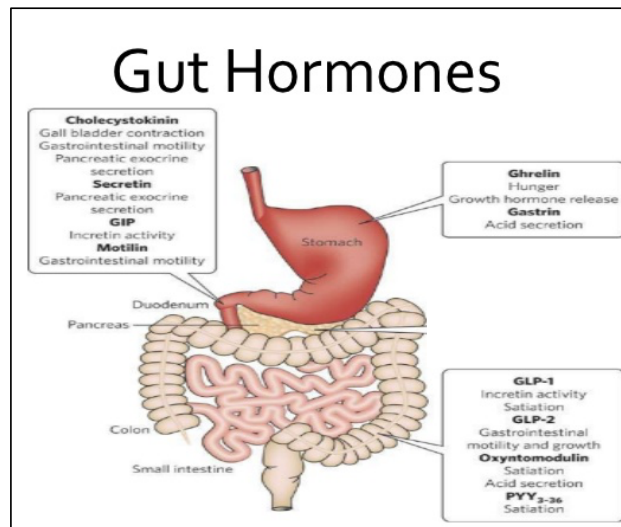


Hormones

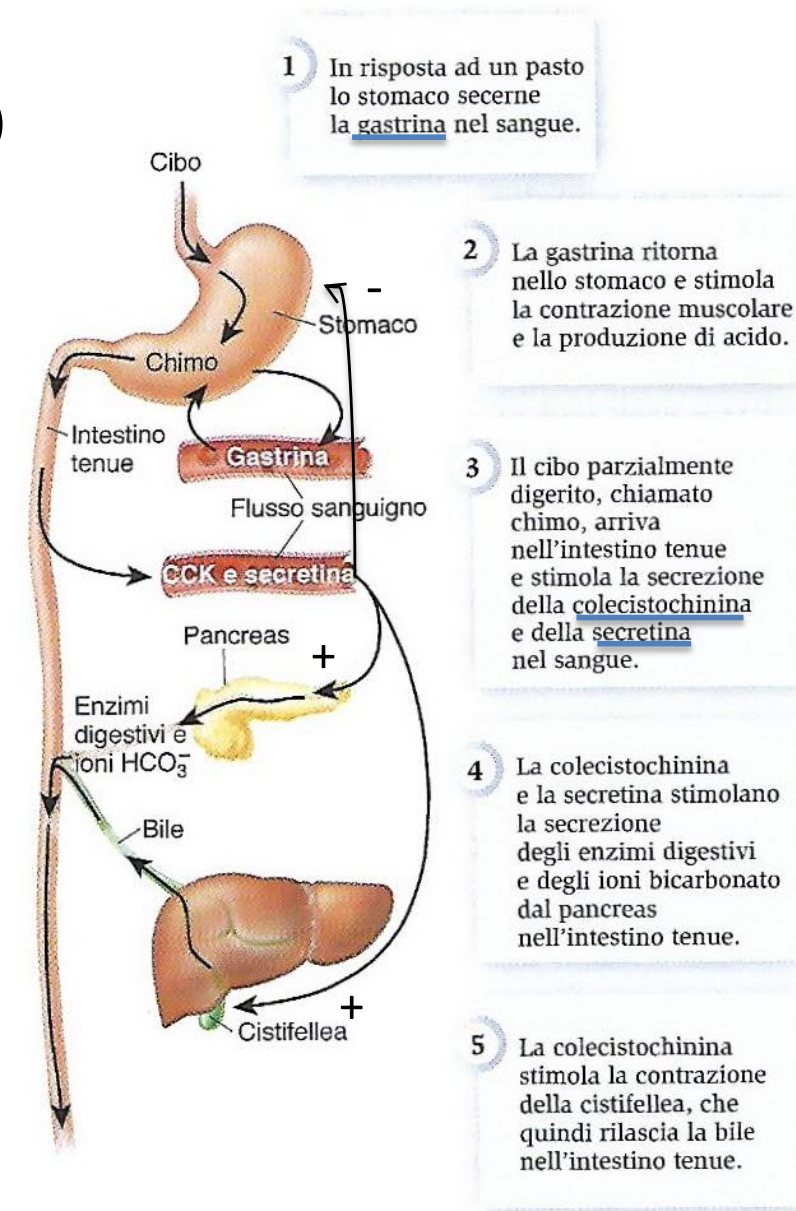


Control of GI tract: hormones

- Secreted by cells (**Enteric Endocrine Cells**) scattered mainly throughout the stomach and small intestine



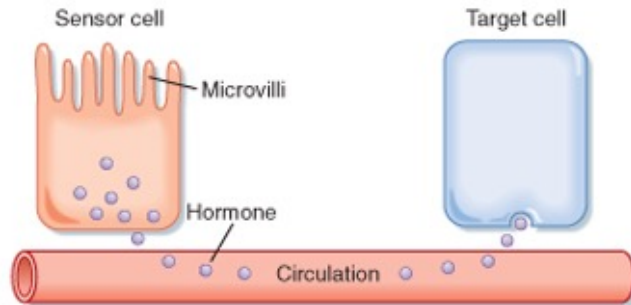
- Hormone's **targets** are:
- Stomach** (gastrin)
- gallbladder and pancreas** (secretin and CCK)



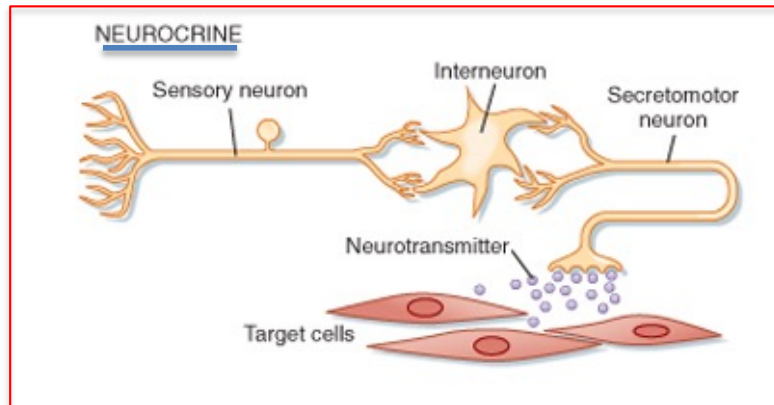
Control of GI tract: innervation

THREE MECHANISMS OF COMMUNICATION MEDIATE RESPONSES IN THE GI TRACT

ENDOCRINE



NEUROCRINE



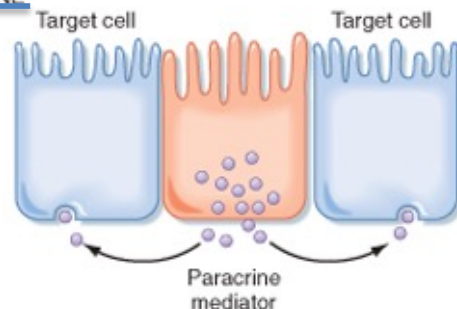
Extrinsic innervation

- Sensory innervation
- Autonomic nervous system
- Motorneuron (esophagus, external sphincter)

Intrinsic innervation

- Enteric nervous system

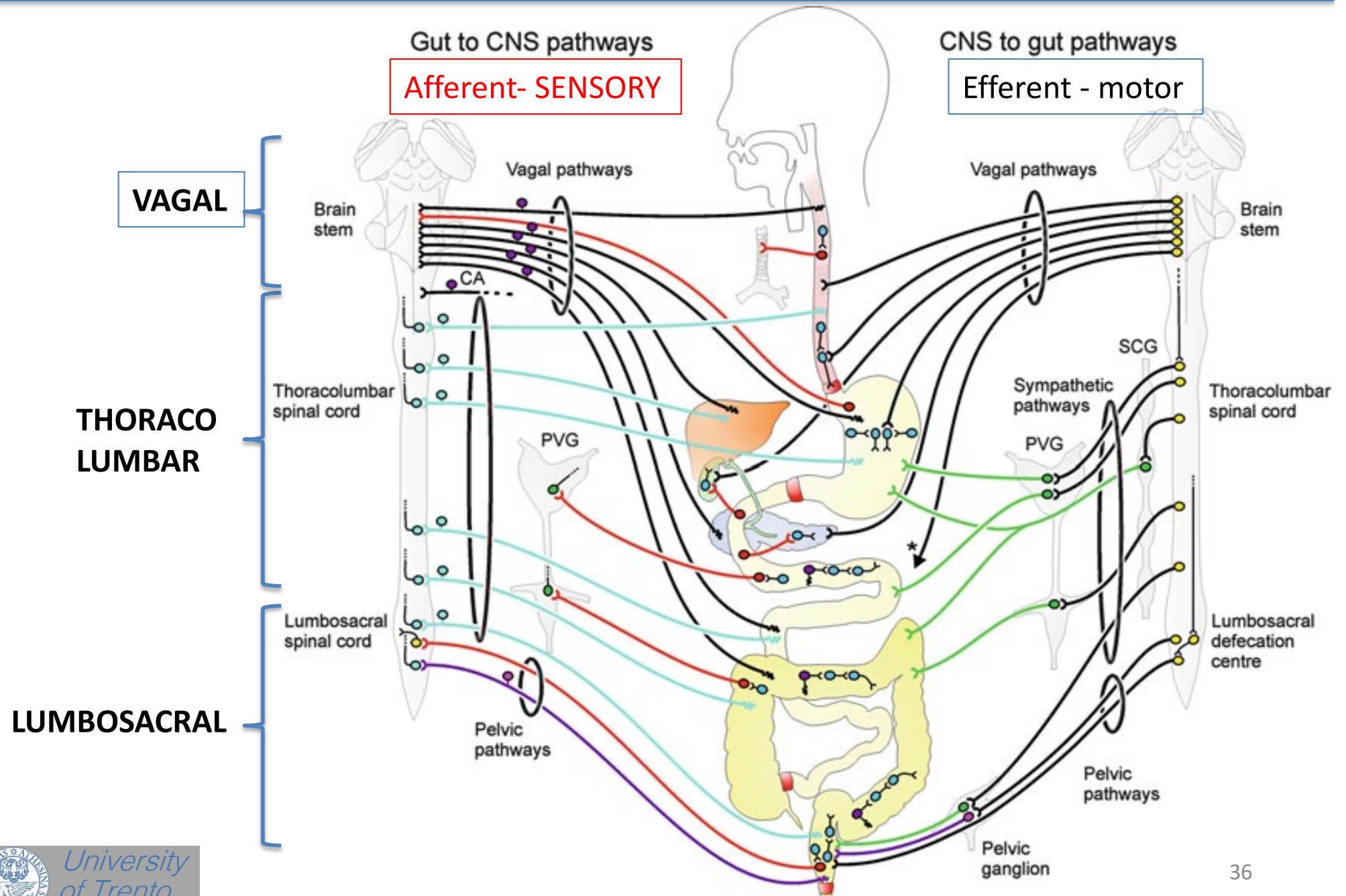
PARACRINE



Integrative control of GI function



Extrinsic Innervation of the gastrointestinal tract

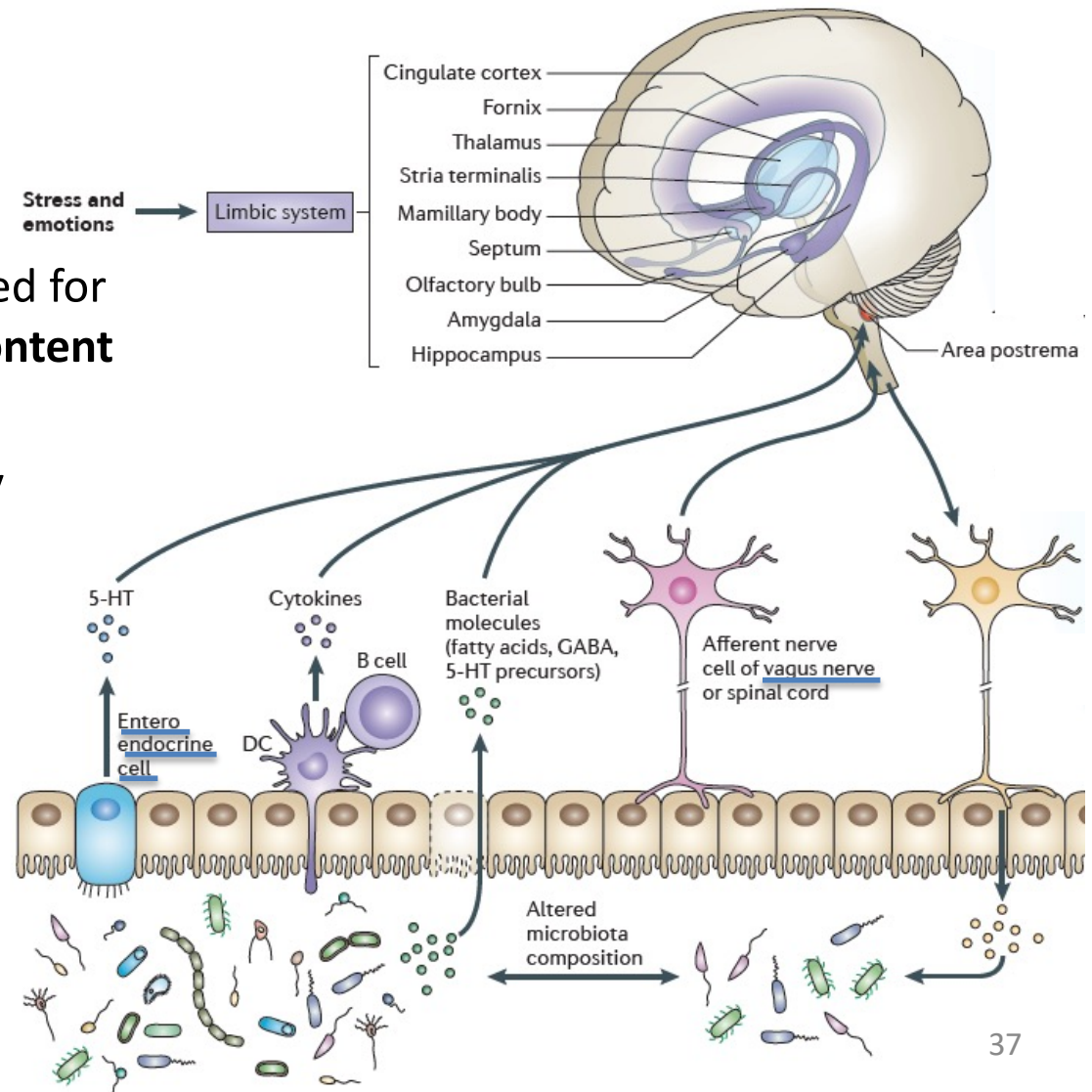


Sensory innervation

- Provide **sensory innervation** especially **from the proximal gut** : esophagus, stomach, proximal small intestine Includes **mechanoreceptors, tension receptors, and chemoreceptors**

- **A similar process** have been proposed for also for the detection of **bacterial content**

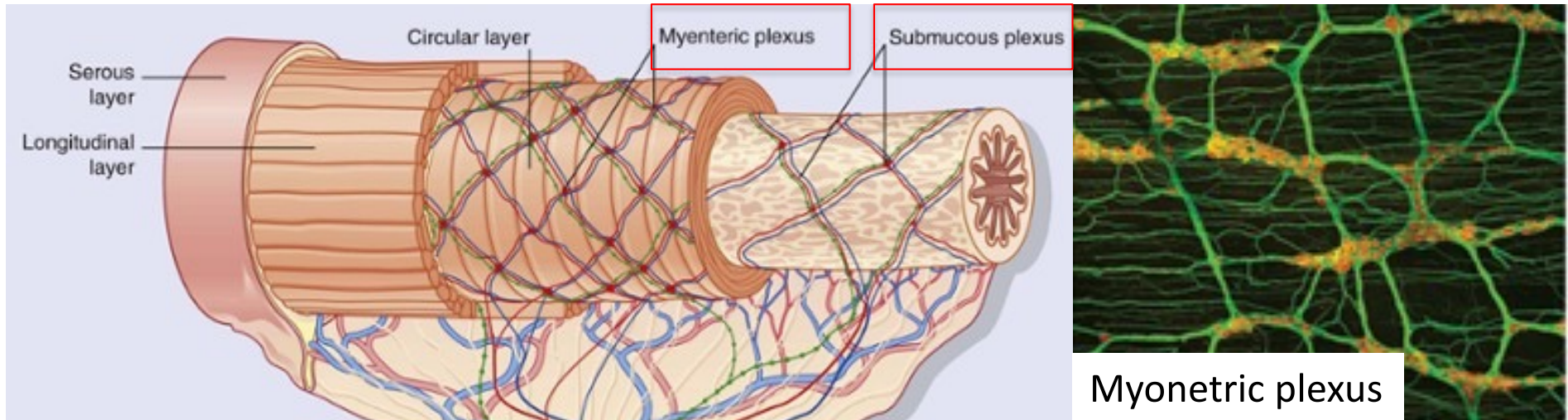
- Alterations in **bacterial content** may induce **changes in the CNS** and changes in the brain can modify the microbiota



Intrinsic Enteric nervous system: A “brain” in the gut



Intrinsic innervation (ENS)



The **enteric nervous system** is formed by:

- **thousands of small ganglia**
- **~200-600 million neurons** in human (> than total sympathetic; ~ all spinal cord neurons)

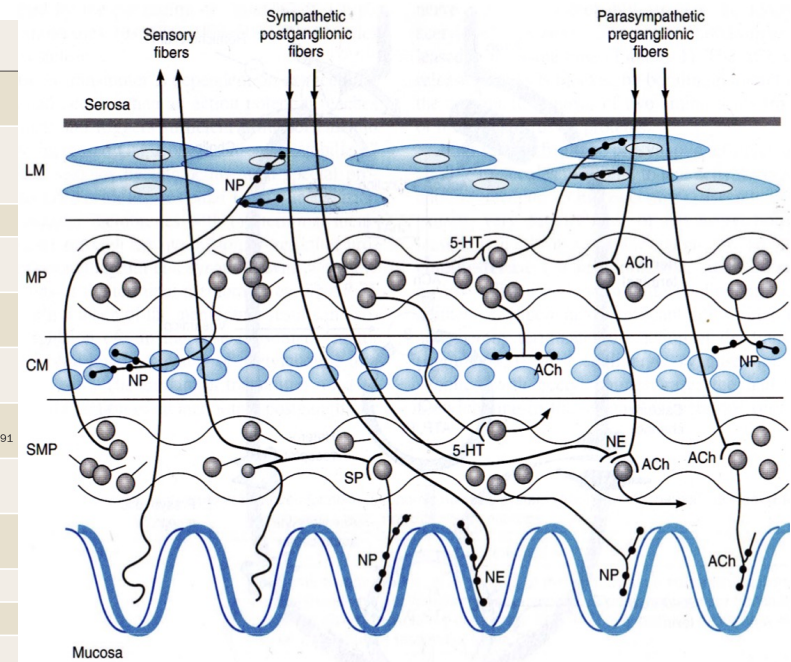
Myenteric plexus is extending from the **upper esophagus to anal sphincter** → **major role in controlling motility**

Submucosal plexus is **absent in the esophagus** and **scarce in the stomach** → mainly associated with district with large fluid fluxes → play their **major role in controlling secretion of water and electrolytes**

Enteric Neuron types

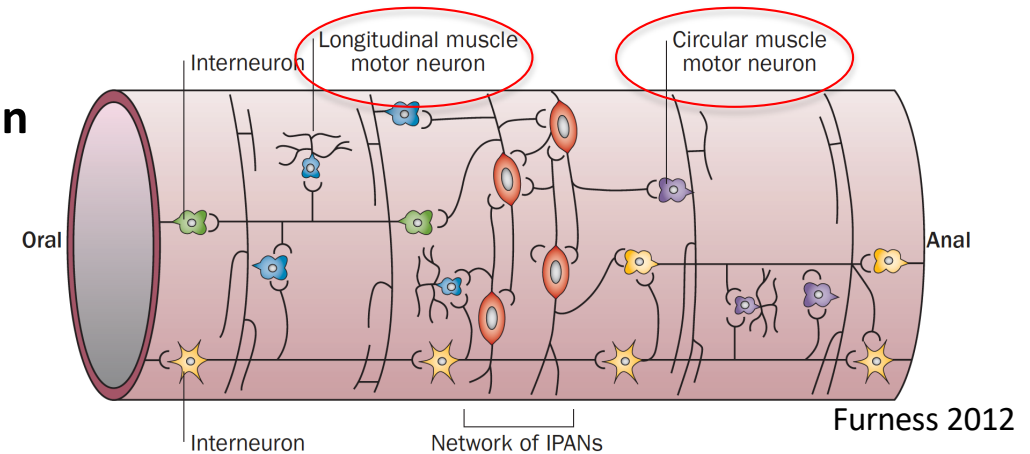
- More than a dozen of **different neuronal types** with different **location**, employed **neurotransmitter**, and **function** have been described in the ENS

Type of neuron	Primary transmitter	Secondary transmitters, modulators	Other neurochemical markers	Study
Enteric excitatory muscle motor neuron	ACh	Tachykinin, enkephalin (presynaptic inhibition)	Calretinin, γ -aminobutyric acid	Brookes <i>et al.</i> (1991); ⁷⁶ Holzer & Holzer Petsche (1997); ⁷⁷ Grider (2003) ⁷⁸
Enteric inhibitory muscle motor neuron	Nitric oxide	VIP, ATP or ATP-like compound, carbon monoxide	PACAP, opioids	Fahrenkrug <i>et al.</i> (1978); ⁷⁹ Costa <i>et al.</i> (1992); ⁸⁰ Sanders & Ward (1992); ⁸¹ Xue <i>et al.</i> (2000) ⁸²
Ascending interneuron	ACh	Tachykinin, ATP	Calretinin, enkephalin	Brookes <i>et al.</i> (1991) ⁸³
ChAT, NOS descending interneuron	ATP, ACh	ND	Nitric oxide, VIP	Young <i>et al.</i> (1995); ⁸⁴ Brookes (2001) ⁸⁵
ChAT, 5-HT descending interneuron	ACh	5-HT, ATP	ND	Furness & Costa (1982); ⁸⁶ Monro <i>et al.</i> (2002); ⁸⁷ Gwynne & Bornstein (2007) ⁸⁸
ChAT, somatostatin descending interneuron	ACh	ND	Somatostatin	Gwynne & Bornstein (2007); ⁸⁸ Portbury <i>et al.</i> (1995) ⁸⁹
Intrinsic sensory neuron	ACh, CGRP, tachykinin	ND	Calbindin, calretinin, IB4 binding	Grider (2003); ⁷⁸ Gwynne & Bornstein (2007); ⁸⁸ Li & Furness (1998); ⁹⁰ Johnson & Bornstein (2004) ⁹¹
Interneurons supplying secretomotor neurons	ACh	ATP, 5-HT	ND	Suprenant (1984); ⁹² Monro <i>et al.</i> (2004) ⁹³
Noncholinergic secretomotor neuron	VIP	PACAP	NPY (in most species)	Cassuto <i>et al.</i> (1981); ⁹⁴ Banks <i>et al.</i> (2005) ⁹⁵
Cholinergic secretomotor neuron	ACh	ND	Calretinin	Brookes <i>et al.</i> (1991); ⁸³ Keast <i>et al.</i> (1985) ⁹⁶
Motor neuron to gastrin cells	GRP, ACh	ND	NPY	Holst <i>et al.</i> (1987); ⁹⁷ Weigert <i>et al.</i> (1997) ⁹⁸
Motor neurons to parietal cells	ACh	Potentially VIP	ND	Nilsson <i>et al.</i> (1972); ⁹⁹ Feldman <i>et al.</i> (1979) ¹⁰⁰



Neural Control of muscle activity in the Small intestine and Colon

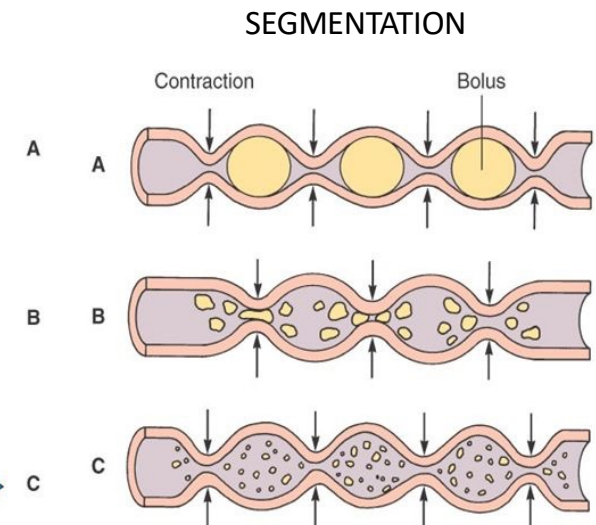
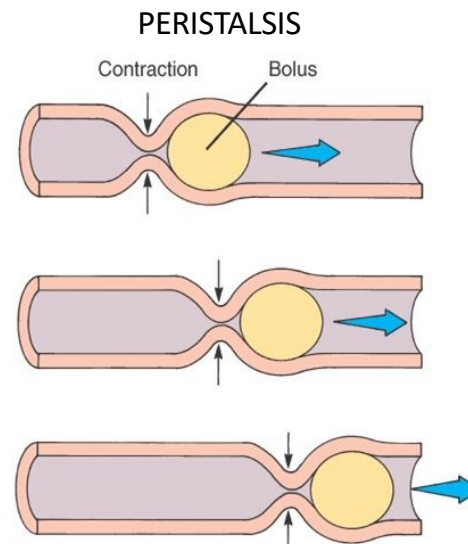
- Movements in the **small intestine and colon** heavily **rely on ENS circuits** innervating both **circular and longitudinal muscle**



- **Different types of movements:**
 - **Peristalsis** (Rapid orthograde propulsion of content)
 - Migrating myoelectric complex (Slow orthograde propulsion)
 - **Segmentation** (mixing movements)
 - **Retropulsion** (vomiting)

- **Largely undefined** are the **mechanism** regulating the **passage** from one pattern of activity to another:

- **Fatty acids** added to the luminal face convert **propulsive** into **segmentation** activity through a **neural mechanism**

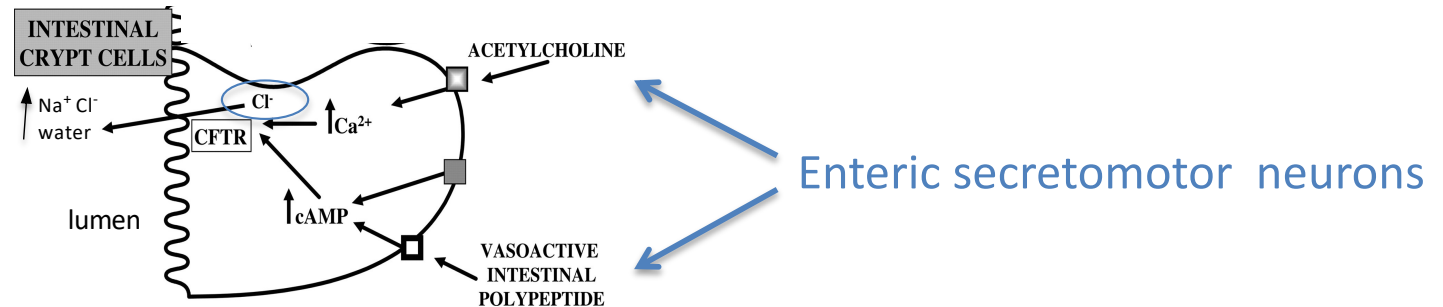


GI tract secretory activity is controlled by neurons

Neurons are mainly involved in 3 secretory functions:

a. **Secretion of Chloride ions in the intestine (accompanied by water/sodium secretion)**

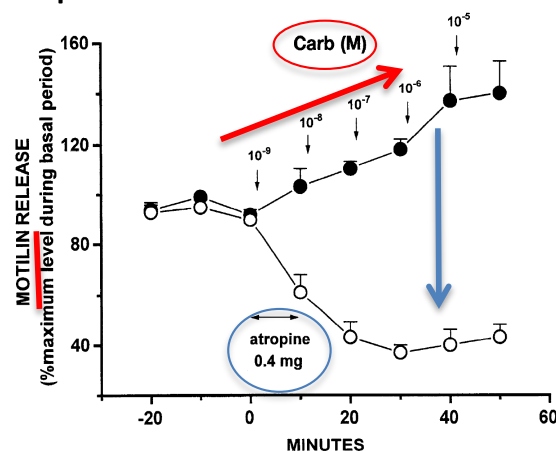
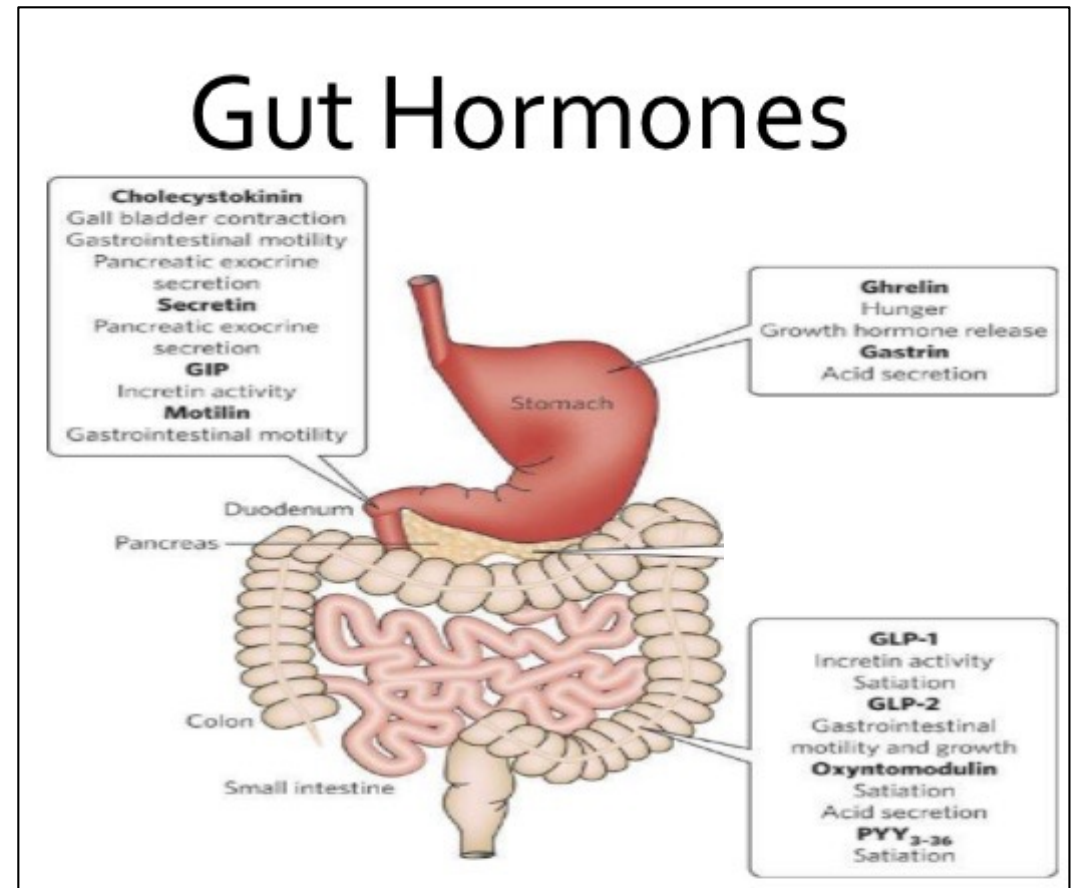
- **Uncontrolled activation** of enteric secretomotor neurons after infection with **Cholera toxin**, pathogenic *E. coli* and rotavirus may cause **severe diarrhea**
(Podewils 2004)



b. **Secretion of gastric acids from gastric parietal cells**

c. Control of function of the mucosal Enteric Endocrine Cells (EEC):

- At least a **dozen different EEC** are **present** in the mucosa of the GI
- For some of them a **functional enteric motor innervation** have been documented:
 - **Gastrin**
 - **PeptideYY**
 - **Motilin**
- **Secretion of PeptideYY and motilin** from EEC **depends on Ach** as they are induced by carbachol and inhibited by atropine



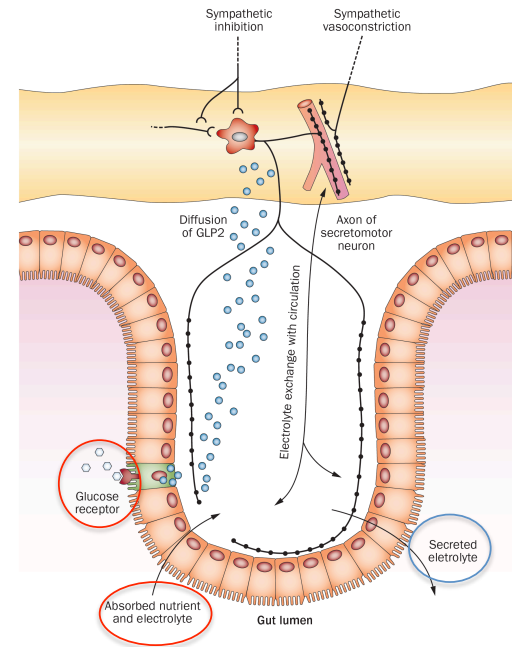
Intestine perfused ex vivo with:

Carbachol: muscarinic agonist
Carbachol+Atropin (antagonist)

Other functions of enteric neurons

- **Influencing glucose transport:**

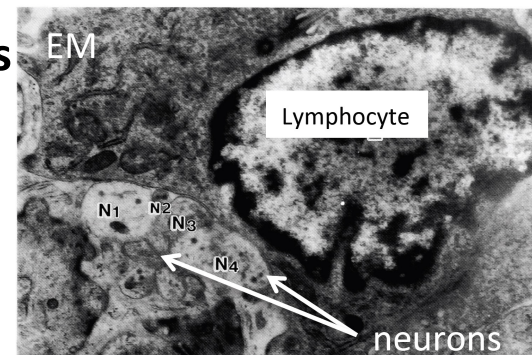
- This class of neurons are innervating the **intestine** are **inducing** the **expression of a glucose transporter in the epithelium** as part of a **reflex that is inducing glucose uptake**
- **Every 100g of adsorbed glucose** also **1.8L of water** is adsorbed



Furness 2012

- **Modulating immune response and Lymphoid tissues such as:**

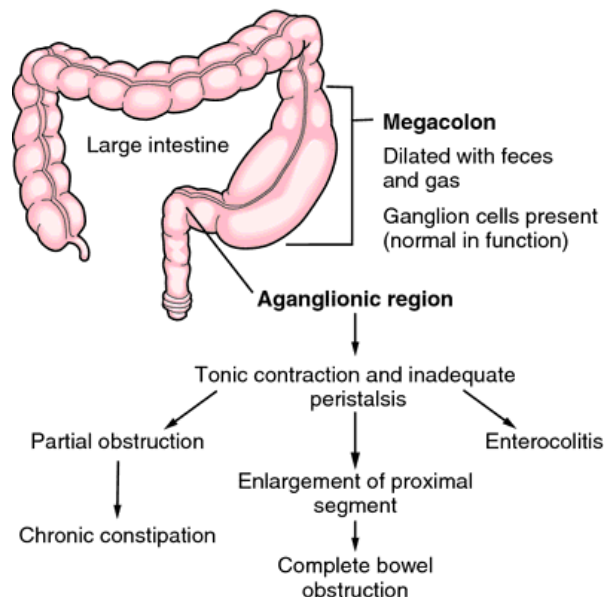
- the Peyer's patches
- Isolated lymphocytes, macrophages and mast cells dispersed in the mucosa



Ichikawa 1992

Enteric neuropathies demonstrate the essential role of the ENS

- Enteric neuropathies demonstrate the **essential role of the ENS**
- For example **Chagas disease** (acquired) or Hirschsprung disease (congenital) have a significant effect on motor function of the digestive tract



Chagas disease

Bites from an insect found in Latin America can introduce *Trypanosoma cruzi* parasites into the human blood stream, which can cause Chagas disease. It can be spread by blood transfusions and organ transplants.

Triatomine insect

The insect can pass Trypanosoma parasites to human hosts when it bites.

T. cruzi parasite

The one-celled parasite feeds on heart and gastrointestinal tissues.



Insect shown to scale, about 3/4 inch long



A million *T. cruzi* could fit inside a single photo pixel.

Sources: National Institutes of Health, University of Texas at Arlington

CHRONIC phase of Chagas Disease

1. Nerve Degeneration
2. Cardiomyopathy (80%)
Heart arrhythmia and blocks
Heart enlargement (cardiomegaly)
Apical aneurism
3. Megaesophagus (25%)
4. Megacolon (30%)

Box 1 | A classification of enteric neuropathies

Congenital or developmental neuropathies

Hirschsprung disease (colorectal aganglionosis)

Hypertrophic pyloric stenosis

Multiple endocrine neoplasia 2B

Neuronal intestinal dysplasia

Mitochondriopathies affecting enteric neurons

Sporadic and acquired neuropathies

Chagas disease

Neurogenic forms of intestinal pseudo-obstruction

Slow transit constipation

Chronic constipation, including constipation of aging

Pathogen-induced diarrhea

IBS

Autoimmune enteric neuritis, paraneoplastic syndrome

Enteric neuritis of unknown etiology

Internal anal sphincter achalasia

Neuropathies secondary to, or associated with, other disease states

Diabetic gastroparesis and other diabetes-related motility disorders

Enteric neuropathy of Parkinson disease

Enteric neuropathy of prion disease

Enteric neuropathies associated with mental retardation or other central nervous system disorders

Ischemic enteric neuropathy, such as in ischemic colitis

Iatrogenic or drug-induced neuropathies

Disorders initiated by antineoplastic drugs (including vinca alkaloids and cisplatin)

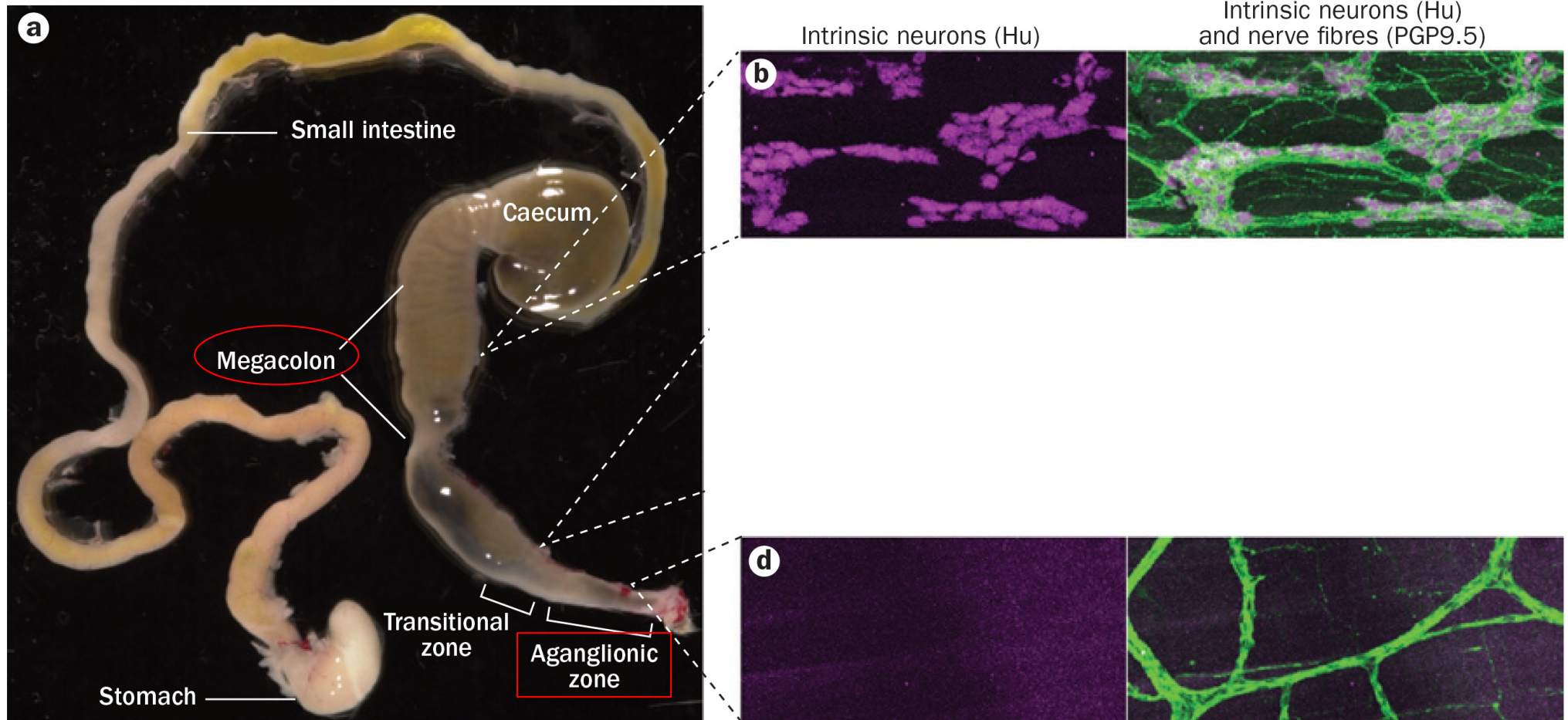
Postoperative ileus

Neuropathy of ischemia, reperfusion injury, such as that associated with intestinal transplantation

Opioid-induced constipation (usually caused when opioids are used to treat chronic pain)

Hirschsprung disease

The Hirschsprung disease distinctive phenotype is **megacolon formation** due to the specific **absence of ENS mediated propulsive activity** in distal bowel (only **extrinsic innervation is present**)



2weeks old **Endothelin (Edn) KO mouse**: a model for Hirschsprung diseases

Obermayer 2013

Hirschsprung disease

- Hirschsprung disease is a **congenital** disease with an overall incidence **1:5000 individuals**
- Mutation in **>12 genes** have been associated with Hirschsprung disease
 - **Short-segment** (80% of patients) → aganglionosis in **rectosigmoid colon**
 - **Long-segment** → aganglionosis extend to substantial part of the colon/intestine
- **Symptoms** include abdominal distension, severe constipation, vomiting and enterocolitis.



- **Treatment** is the surgical removal of the aganglionic region

Summary

- All organisms require nutrients to survive
 - organic (carbohydrates, proteins, lipids, nuclei acids, vitamins) or inorganic (minerals, water)
 - some are essential
- The gastro-intestinal tract is organized in functional regions specialized in specific process
 - Digestion begins in mouth
 - Protein digestion continues in stomach
 - Most of the digestion and absorption is occurring in small intestine
 - In large intestine fecal concentration and storage is occurring
- Accessory organs (liver/gallbladder) and pancreas contribute to the process of digestion
- Bacteria contribute to the process and produce some essential vitamins
- The function of gastrointestinal tract is controlled mainly by:
 - Hormones released by entero endocrine cells
 - Extrinsic (autonomic) and intrinsic (ENS) nervous system

Bibliography

Zoologia

Hickman, Roberts, Keen, Eisenhour, Larson, L'Anson
McGraw-Hill; 2016.

Chapters: 7.

Biologia 5 Animali

Brooker, Widmaier, Graham
McGraw-Hill; 2011.

Chapters: 41.

In-depth studies:

The enteric nervous system and neurogastroenterology

John B. Furness

NATURE REVIEWS | GASTROENTEROLOGY & HEPATOLOGY VOLUME 9 | MAY
2012 | **286**